

Mostly Harmless Heterogeneous Macro

A Guided tour to Theoretical Macroeconomics

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Abstract

This note is a practical manual for closed-economy New Keynesian models with increasing degrees of household heterogeneity. It starts from the representative-agent benchmark, moves to a two-agent New Keynesian economy, then introduces a tractable heterogeneous-agent intermediate step—THANK—and culminates in a baseline HANK model with incomplete markets and idiosyncratic labor-income risk. **The aim is strictly pedagogical.** Every environment is presented with household and firm problems, first-order conditions, equilibrium restrictions, nominal rigidities, monetary and fiscal rules, market-clearing conditions, and computational logic. The exposition is intentionally slow. Definitions, timing conventions, log-linearization, and the role of firms, profits, fiscal incidence, and redistribution are all developed in detail so that students can move from intuition to equations and then to code without losing the thread.

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1 Introduction

This handbook is written as a guided route through modern New Keynesian macroeconomics. I strongly encourage the reader to study Florin Bilbiie's handbook (available here: <https://sites.google.com/site/florinbilbiie/thank-book>), one of the leading macroeconomic researchers

during the last decades. It is neither a substitute for standard references such as Woodford [13] or Galí [10], but a complement.

The point is narrower and more practical: to provide a route through the main families of New Keynesian models with household heterogeneity while keeping the exposition slow enough that a graduate student can see where each equation comes from, what each block of the model does, and why each extra layer of heterogeneity matters.

The strategy is simple. We hold the nominal side of the model as fixed as possible and gradually enrich the household side. The point is not to present four unrelated models. The point is to show that RANK, TANK, THANK, and HANK are best understood as different versions of one broad architecture, with progressively richer internal heterogeneity and progressively higher computational demands.

The progression matters for several reasons.

- (i) It clarifies mechanisms. Many aggregate impulse responses that students first see in representative-agent New Keynesian models survive in richer heterogeneous-agent versions. What changes is often not the sign of the aggregate response, but its internal composition across households.
- (ii) It clarifies interpretation. Once heterogeneity is introduced, monetary policy is no longer only about intertemporal substitution. It also works through disposable income, profits, taxes, debt ownership, liquidity positions, transfers, and redistribution across households with different marginal propensities to consume.
- (iii) It clarifies model choice. A student should know not only how to solve a model, but also when each model is the right tool.
- (iv) It clarifies computation. RANK is analytically neat and numerically trivial. TANK remains almost as simple. THANK is still analytically tractable in an important sense, but reproduces key channels of HANK. HANK, by contrast, requires dynamic programming, policy functions, and the forward evolution of a cross-sectional distribution.

The literature motivates this progression. RANK remains the benchmark environment for textbook New Keynesian macroeconomics. TANK-type environments are closely related to the literature on rule-of-thumb or hand-to-mouth households, such as Campbell and Mankiw [8], Bilbiie [4], and Galí et al. [11]. THANK, in the sense emphasized by Bilbiie [6], is a tractable HANK framework that analytically captures several of the micro-heterogeneity channels that matter in quantitative HANK, including cyclical inequality, risk, self-insurance, precautionary saving, and realistic intertemporal marginal propensities to consume. HANK models grow out of the incomplete-markets tradition of Aiyagari [1], brought into New Keynesian analysis by papers such as Kaplan et al. [12] and interpreted through redistribution channels by Auclert [2]. On the computational side, modern sequence-space methods have greatly expanded what is feasible in heterogeneous-agent macro [3].

This manual is organized as follows. Section 2 explains where these models came from, why they were needed, and what criticisms have shaped them. Section 3 explains how to read these models without getting lost. Section 4 reviews the mathematical toolkit used throughout. Section 6 gives a conceptual roadmap from RANK to TANK to THANK to HANK. Section 7 presents the common

New Keynesian environment. Section 8 develops the firm side in detail. Section 5 explains what log-linearization is and why it is useful. Sections 9, 10, 11, and 12 present the four household blocks. Section 13 compares them. Section 14 sketches the computational logic. Section 15 enriches the framework with a richer fiscal block and a more structured labor market with unions and sticky wages.

2 Historical background, motivation, and criticisms

This manual studies RANK, TANK, THANK, and HANK as a sequence of increasingly rich New Keynesian environments. Before turning to technical details, it is useful to explain why these models exist, what historical problems they were designed to solve, and which criticisms shaped their evolution.

2.1 From old Keynesianism to the New Keynesian synthesis

The intellectual background of New Keynesian macroeconomics lies in the attempt to combine three ideas that were once often presented separately.

- (i) The Keynesian claim that nominal and real rigidities matter for short-run fluctuations.
- (ii) The rational-expectations and intertemporal-optimization revolution of the 1970s and 1980s.
- (iii) The monetary-policy focus on interest-rate rules, credibility, and inflation stabilization.

The old Keynesian tradition emphasized aggregate demand failures, sticky prices, and unemployment, but often relied on ad hoc behavioral relations and static or semi-static frameworks. The rational-expectations critique then pushed macroeconomics toward models with explicit optimization and internally consistent expectations. The resulting challenge was clear: could one build models with fully optimizing agents and rational expectations that still delivered a meaningful role for monetary policy and nominal rigidities?

The modern New Keynesian framework is the answer to that question. The central synthesis was to embed monopolistic competition and nominal rigidities into otherwise standard intertemporal general-equilibrium models. In that synthesis, households optimize, firms optimize, expectations are forward-looking, and monetary policy affects the economy because prices or wages do not adjust frictionlessly. The resulting framework is the one codified and systematized most famously by Woodford [13] and presented pedagogically by Galí [10].

2.2 Why RANK was the natural starting point

The first fully successful New Keynesian benchmark was representative-agent. This was not because researchers believed the world literally contains one household. It was because the representative-agent benchmark isolates the aggregate logic of nominal rigidities and interest-rate policy as cleanly as possible.

RANK became the workhorse for at least four reasons.

- (i) It delivered a transparent dynamic IS relation from household optimization.

- (ii) It delivered a Phillips curve from forward-looking sticky-price pricing.
- (iii) It allowed monetary policy to be represented by an interest-rate rule rather than by money supply growth alone.
- (iv) It generated a compact equilibrium system suitable for theory, welfare analysis, and policy design.

For teaching, this was a major victory. One could explain inflation dynamics, monetary transmission, optimal policy, and determinacy in a unified, fully microfounded framework.

2.3 Why RANK eventually became unsatisfactory

As empirical work on household heterogeneity, consumption behavior, and distribution gained importance, the limitations of the representative-agent benchmark became increasingly visible.

The main problems were the following.

- (i) Households are heterogeneous in wealth, income risk, and access to liquidity.
- (ii) Many households appear to have high marginal propensities to consume out of current income.
- (iii) The distribution of taxes, transfers, profits, and debt ownership matters for aggregate demand.
- (iv) Monetary policy affects different households through different channels, not only through one common Euler equation.

In a representative-agent model, all households respond to the real interest rate in the same way. In reality, many households do not behave as if they are smoothly solving a frictionless intertemporal consumption problem with full asset-market access. This is the empirical and conceptual pressure that pushed the literature toward models with richer household sectors.

2.4 The rise of TANK

The simplest response to the limitations of RANK was not HANK. It was TANK. The idea was to preserve as much of the tractability of representative-agent macro as possible while acknowledging that not all households smooth consumption intertemporally.

The two key inspirations were empirical and conceptual.

- (i) Empirically, consumption appears to track current income much more closely for some households than the representative-agent model suggests.
- (ii) Conceptually, fiscal and monetary policy can have larger effects if part of the household sector reacts strongly to current disposable income.

This gave rise to models with one Ricardian household type and one non-Ricardian or hand-to-mouth type. The classic references include Campbell and Mankiw [8], Bilbiie [4], and Galí et al. [11].

2.5 What TANK got right

TANK was an important step forward because it clarified several points.

- (i) Heterogeneity can matter even in the simplest possible way.
- (ii) Fiscal incidence matters. A transfer to a high-MPC household is not equivalent to a transfer to a low-MPC household.
- (iii) Monetary policy no longer works only through one Euler equation. Part of demand becomes current-income driven.

2.6 Why TANK was still not enough

TANK is deliberately thin. It captures the distinction between hand-to-mouth and Ricardian behavior, but it still suppresses several central ingredients of the household side.

- (i) It does not generate an endogenous wealth distribution.
- (ii) It does not model precautionary saving.
- (iii) It does not allow households to transition across a rich state space.
- (iv) It cannot reproduce a realistic cross-sectional distribution of MPCs.
- (v) It does not fully capture how policy affects risk and self-insurance.

These limitations matter if the research question concerns distributional dynamics, not just the existence of two household groups.

2.7 The rise of HANK

HANK models emerged from the attempt to combine incomplete-markets macroeconomics with New Keynesian nominal rigidities. The intellectual background is the incomplete-markets tradition of Aiyagari [1]. The key innovation of HANK was to place that tradition inside a New Keynesian environment with sticky prices, policy rules, and aggregate demand management.

The payoff was immediate. HANK makes it possible to study how monetary and fiscal policy interact with:

- (i) heterogeneity in liquid and illiquid wealth;
- (ii) uninsurable idiosyncratic labor-income risk;
- (iii) borrowing constraints;
- (iv) endogenous precautionary saving;
- (v) redistribution across households with very different MPCs.

Kaplan et al. [12] is a defining reference in this tradition. Auclert [2] then sharpened the redistribution perspective by decomposing the effects of monetary policy into several channels, including earnings heterogeneity and Fisher-type redistribution.

2.8 Why HANK also created a new problem

HANK solved many conceptual problems, but created a new one: tractability. Once the household sector is summarized by a full policy-function problem and a cross-sectional distribution, the model becomes much harder to solve, explain, and use in analytical work.

This is the main reason why the literature continued to search for intermediate environments. If one wants the main intuition of HANK without immediately moving to a high-dimensional computational problem, one needs something between TANK and HANK.

2.9 Why THANK is useful

THANK fills exactly that role. In the Bilbiie tradition, THANK is not just another TANK. It is a tractable HANK. The aim is to capture analytically several channels that are central in HANK:

- (i) cyclical inequality;
- (ii) transition risk;
- (iii) self-insurance;
- (iv) precautionary saving;
- (v) realistic iMPCs.

This makes THANK especially useful pedagogically. It lets students see how one moves from static type heterogeneity to risk-sensitive, state-contingent heterogeneity before confronting the full machinery of HANK.

3 How to read these models without getting lost

Many students first encounter heterogeneous-agent macroeconomics as an intimidating sequence of equations. The difficulty is often not that any single equation is hard. The difficulty is that it becomes unclear which equations are behavioral, which are equilibrium conditions, which are identities, and which are approximations. A useful first step is therefore to separate the model into blocks.

3.1 Five blocks

For almost all models in this manual, the equilibrium can be decomposed into five blocks.

- (i) The household block. This describes preferences, budget constraints, Euler equations, borrowing constraints, labor-supply conditions, and, in HANK, policy functions and the law of motion of the distribution.
- (ii) The firm block. This describes production, cost minimization, markup pricing, sticky-price pricing, and profit distribution.
- (iii) The policy block. This describes monetary and fiscal policy.

- (iv) The aggregation block. This maps individual or group choices into aggregate variables.
- (v) The market-clearing block. This closes the model.

Whenever a student sees a new equation, the first question should be: *to which block does this equation belong?*

3.2 Three representations of the same model

A second source of confusion is that the same model often appears in three distinct representations.

- (i) A micro representation, in terms of household and firm optimization problems.
- (ii) An equilibrium representation, in terms of first-order conditions, constraints, and market-clearing restrictions.
- (iii) An approximate aggregate representation, usually log-linear, in terms of a few equations such as an IS curve and a Phillips curve.

The best route through the material is therefore

optimization problem $\longrightarrow$ equilibrium system $\longrightarrow$ approximate aggregate system.

3.3 Why this manual is long

The slow route is often the faster route in the long run. Many misunderstandings in macro come from manipulating reduced-form equations without understanding where they came from. A long manual is useful if it prevents those misunderstandings from appearing at the foundation.

4 Mathematical toolkit

This section collects a few mathematical ideas that will be used repeatedly.

4.1 Expected utility

All household problems in this manual are instances of expected discounted utility maximization:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t, N_t, \dots), \quad (1)$$

where $\beta \in (0, 1)$ is the discount factor. The expectation operator appears because households or firms care about future states that may be uncertain.

4.2 Lagrangians

A Lagrangian is a device for solving constrained optimization problems. In dynamic settings, one generally imposes the flow budget constraint period by period and associates a multiplier with each date. The multiplier is often interpretable as the marginal utility value of one additional unit of wealth.

4.3 Bellman equations

In dynamic programming, instead of choosing an entire infinite sequence at once, the household solves recursively. A Bellman equation has the form

$$V(x) = \max_{x'} \{u(x, x') + \beta \mathbb{E}[V(x')]\}. \quad (2)$$

4.4 Markov transitions

Many heterogeneous-agent models describe households as moving across a finite set of states over time. A convenient way to represent those movements is with a Markov chain.

A process is Markov if the probability of tomorrow's state depends only on today's state, not on the full earlier history. If the possible states are indexed by $i, j \in \{1, \dots, n\}$, the transition matrix is

$$\Pi = \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix}, \quad (3)$$

where

$$p_{ij} = \Pr(s_{t+1} = j \mid s_t = i), \quad (4)$$

and each row sums to one:

$$\sum_{j=1}^n p_{ij} = 1 \quad \text{for all } i. \quad (5)$$

In the simplest two-state case,

$$\Pi_t = \begin{pmatrix} p_{SS,t} & p_{SH,t} \\ p_{HS,t} & p_{HH,t} \end{pmatrix}, \quad (6)$$

where, for example, $p_{SH,t}$ is the probability that a household currently in state S moves to state H next period.

If μ_t denotes the row vector of population shares across states at date t , then the distribution evolves according to

$$\mu_{t+1} = \mu_t \Pi_t. \quad (7)$$

In a time-invariant environment, a stationary distribution $\bar{\mu}$ satisfies

$$\bar{\mu} = \bar{\mu} \Pi. \quad (8)$$

This notation is especially useful in THANK and HANK. In THANK, it describes transitions between a saver state and a constrained state. In HANK, the same logic generalizes to richer income or employment processes.

4.5 Steady states

A steady state is an allocation in which aggregate quantities are constant over time and individual or group distributions are stationary. In deterministic steady state, all expectations are realized

trivially because the future is identical to the present.

4.6 Log deviations

Given a variable X_t with steady-state value X , define

$$\hat{x}_t \equiv \log \left(\frac{X_t}{X} \right). \quad (9)$$

For small deviations from steady state,

$$\hat{x}_t \approx \frac{X_t - X}{X}. \quad (10)$$

4.7 State variables

A state variable is any variable whose current value matters for future decisions.

- (i) In RANK, aggregate wealth is often the relevant state variable.
- (ii) In TANK, household type matters.
- (iii) In THANK, household type and transition risk matter.
- (iv) In HANK, the entire cross-sectional distribution matters.

5 What does it mean to log-linearize?

Students are often told that a model is *log-linearized around its steady state*. The phrase sounds technical, but the underlying idea is simple. A nonlinear equilibrium system is usually too complicated to solve analytically in levels. If one is interested in small fluctuations around a steady state, however, one can approximate the nonlinear equations by a first-order expansion. The result is a system of linear equations in suitably defined deviations from steady state.

This section explains what that means, why it is useful, and what its limits are.

5.1 Why approximation is needed

Most modern macroeconomic models are nonlinear. Utility is nonlinear, production may be nonlinear, pricing equations are nonlinear, and expectations often enter multiplicatively with endogenous variables. Even when each block of the model is conceptually simple, the full equilibrium system can be difficult to solve exactly.

A first-order approximation is attractive because it turns a nonlinear dynamic system into a linear one. Linear systems are much easier to manipulate, solve, simulate, and interpret. This is the reason why log-linearized representations play such a central role in graduate macroeconomics, especially in the New Keynesian tradition.

The basic logic is local. One does not claim that the economy is literally linear. One claims only that, for sufficiently small deviations around a reference point, the nonlinear system can be well approximated by its tangent.

5.2 The steady state as the reference point

To approximate a model, one first needs a point around which to approximate it. In this manual, that point is usually a deterministic steady state.

A steady state is a configuration in which aggregate variables are constant over time, and in heterogeneous-agent models the relevant cross-sectional distribution is stationary. If there is no trend growth and no time-varying exogenous disturbance, the steady state satisfies the model's equilibrium conditions with

$$X_t = X_{t+1} = X$$

for all aggregate variables of interest.

The steady state matters because it provides the benchmark relative to which fluctuations are measured. When we write a hat variable such as $\hat{x}_t$, we are usually measuring the variable at date t relative to its steady-state value.

5.3 Linearization versus log-linearization

It is useful to distinguish two related ideas.

Linearization in levels. If a variable X_t has steady-state value X , one can define the level deviation

$$\tilde{X}_t \equiv X_t - X.$$

A first-order Taylor expansion in these deviations is a *linearization in levels*.

Log-linearization. If $X_t > 0$, one can instead define the log deviation

$$\hat{x}_t \equiv \log \left(\frac{X_t}{X} \right) = \log X_t - \log X. \quad (11)$$

A first-order expansion expressed in these log deviations is a *log-linearization*.

For small deviations,

$$\hat{x}_t \approx \frac{X_t - X}{X}. \quad (12)$$

Thus a hat variable is approximately a percentage deviation from steady state. This is one reason log-linearization is so convenient: the resulting coefficients are often easy to interpret economically.

In practice, macroeconomists often say “linearization” and “log-linearization” somewhat loosely. But the distinction matters. Additive objects such as net interest rates or inflation rates are often linearized directly in levels, while positive quantities such as output, consumption, wages, prices, or capital are often log-linearized.

5.4 General principle

Suppose the model contains a nonlinear equilibrium condition

$$F(X_t, X_{t+1}, Z_t) = 0, \quad (13)$$

where X_t is an endogenous variable, X_{t+1} is its future value, and Z_t is an exogenous disturbance or state variable.

Let (X, Z) denote a steady state satisfying

$$F(X, X, Z) = 0. \quad (14)$$

Then, for small deviations around that point, a first-order Taylor expansion gives

$$0 \approx F(X, X, Z) + F_1(X, X, Z)(X_t - X) + F_2(X, X, Z)(X_{t+1} - X) + F_3(X, X, Z)(Z_t - Z), \quad (15)$$

where F_1 , F_2 , and F_3 denote partial derivatives of F with respect to its first, second, and third arguments.

Because the steady state satisfies (14), the constant term vanishes. What remains is linear in the deviations:

$$a(X_t - X) + b(X_{t+1} - X) + c(Z_t - Z) \approx 0, \quad (16)$$

for coefficients a , b , and c determined by the derivatives of F at the steady state.

If one rewrites these deviations in log form when appropriate, one obtains the familiar log-linear equations used throughout New Keynesian analysis.

5.5 Why logs are especially convenient

There are three practical reasons why logs are used so often.

First, log deviations are approximately percentage deviations. This makes the size of fluctuations immediately interpretable.

Second, products become sums to a first-order approximation. If

$$Y_t = A_t N_t,$$

then the log-linear approximation becomes

$$\hat{y}_t = \hat{a}_t + \hat{n}_t.$$

This is much easier to interpret than a nonlinear multiplicative relation in levels.

Third, many macroeconomic equilibrium conditions are naturally written in ratios, growth rates, or proportional changes. Logs make those structures more transparent.

5.6 Useful first-order approximation rules

A great deal of log-linearization can be done almost mechanically once a few approximation rules are understood. Let X_t and Q_t be positive variables with steady states X and Q .

Rule 1: logs and percentage deviations.

$$\hat{x}_t = \log\left(\frac{X_t}{X}\right) \approx \frac{X_t - X}{X}.$$

Rule 2: products. If

$$Y_t = X_t Q_t,$$

then

$$\hat{y}_t \approx \hat{x}_t + \hat{q}_t. \quad (17)$$

Rule 3: ratios. If

$$Y_t = \frac{X_t}{Q_t},$$

then

$$\hat{y}_t \approx \hat{x}_t - \hat{q}_t. \quad (18)$$

Rule 4: powers. If

$$Y_t = X_t^\alpha,$$

then

$$\hat{y}_t \approx \alpha \hat{x}_t. \quad (19)$$

Rule 5: sums. Sums are less convenient. In general,

$$Y_t = X_t + Q_t$$

does not imply $\hat{y}_t = \hat{x}_t + \hat{q}_t$. Instead, one must linearize in levels and then normalize by the steady state:

$$\hat{y}_t \approx \frac{X}{Y} \hat{x}_t + \frac{Q}{Y} \hat{q}_t, \quad (20)$$

where $Y = X + Q$ in steady state.

This last rule is especially important in heterogeneous-agent models, where aggregates are often sums across groups or states.

5.7 Example: production

Consider the production function

$$Y_t = A_t N_t. \quad (21)$$

Let the steady state satisfy

$$Y = AN.$$

Taking logs,

$$\log Y_t = \log A_t + \log N_t.$$

Subtract steady-state logs:

$$\log Y_t - \log Y = (\log A_t - \log A) + (\log N_t - \log N).$$

Using hat notation,

$$\hat{y}_t = \hat{a}_t + \hat{n}_t. \quad (22)$$

This example is especially transparent because the relation is exactly log-linear, not merely approximately so. Whenever the original equation is multiplicative, the log transformation often produces an exact linear relation in logs.

5.8 Example: Euler equation

Now consider a simple Euler equation,

$$C_t^{-\sigma} = \beta R_t \mathbb{E}_t [C_{t+1}^{-\sigma}]. \quad (23)$$

To avoid unnecessary clutter, suppose the certainty-equivalent version can be written as

$$C_t^{-\sigma} = \beta R_t C_{t+1}^{-\sigma}. \quad (24)$$

Taking logs gives

$$-\sigma \log C_t = \log \beta + \log R_t - \sigma \log C_{t+1}. \quad (25)$$

In steady state,

$$-\sigma \log C = \log \beta + \log R - \sigma \log C, \quad (26)$$

which implies

$$\log \beta + \log R = 0.$$

Subtracting the steady-state relation from the time- t equation yields

$$-\sigma \hat{c}_t = \hat{R}_t - \sigma \hat{c}_{t+1}. \quad (27)$$

Rearranging,

$$\hat{c}_t = \hat{c}_{t+1} - \frac{1}{\sigma} \hat{R}_t. \quad (28)$$

With uncertainty and rational expectations, this becomes

$$\hat{c}_t = \mathbb{E}_t[\hat{c}_{t+1}] - \frac{1}{\sigma} \hat{R}_t. \quad (29)$$

This is the familiar linearized intertemporal condition: current consumption rises when expected future consumption rises, and falls when the real return to postponing consumption rises.

Strictly speaking, in New Keynesian applications one usually works with the *real* interest rate rather than the nominal rate alone, so expected inflation will enter the linearized Euler equation. But the simplified expression above is useful for understanding the basic mechanics of approximation.

5.9 Example: Taylor rule

Consider the gross-interest-rate Taylor rule

$$1 + i_t = (1 + i)(1 + \pi_t)^{\phi_\pi} \left(\frac{Y_t}{Y} \right)^{\phi_y} e^{\nu_t}, \quad (30)$$

where ν_t is a monetary-policy shock.

Take logs:

$$\log(1 + i_t) = \log(1 + i) + \phi_\pi \log(1 + \pi_t) + \phi_y \log\left(\frac{Y_t}{Y}\right) + \nu_t. \quad (31)$$

Around a zero-inflation steady state and for small inflation,

$$\log(1 + \pi_t) \approx \pi_t.$$

Also,

$$\log\left(\frac{Y_t}{Y}\right) = \hat{y}_t.$$

Define the log deviation of the gross nominal interest rate by

$$\hat{i}_t \equiv \log\left(\frac{1 + i_t}{1 + i}\right).$$

Then the first-order approximation becomes

$$\hat{i}_t = \phi_\pi \pi_t + \phi_y \hat{y}_t + \nu_t. \quad (32)$$

This is the standard linear policy rule used in many textbook New Keynesian systems.

5.10 Example: the New Keynesian Phillips curve

A useful way to understand the role of log-linearization is to look at the price-setting block. The nonlinear Calvo pricing system is not especially transparent in levels. But once one log-linearizes the reset-price equation and the aggregate price-index equation around a zero-inflation steady state, the pricing block collapses into the familiar New Keynesian Phillips curve

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa \hat{m}c_t. \quad (33)$$

This is a good example of what approximation achieves. The underlying pricing problem is dynamic, nonlinear, and involves a distribution of prices across firms. The log-linearized representation condenses all that structure into a single equation linking current inflation to expected future inflation and real marginal cost.

5.11 A note on certainty equivalence and what survives at first order

First-order approximations eliminate many terms that matter only at second order or higher. In particular, risk-adjustment terms, precautionary motives, volatility effects, and many interactions involving cross-sectional dispersion often disappear from a first-order system.

This is why first-order perturbation methods are often associated with a kind of certainty equivalence: the linearized dynamics of averages are governed by expected future values, but not by higher moments such as variances, unless those moments enter already at first order.

This point is especially important in heterogeneous-agent macroeconomics. In RANK, many core insights survive first-order approximation quite well. In TANK, some heterogeneity survives because group-level aggregation enters already at first order. But in THANK and especially HANK, some of the most interesting mechanisms — precautionary saving, changing risk exposure, nonlinear

borrowing constraints, distributional tails, and endogenous movements in the distribution — are not well captured by a purely first-order local approximation.

5.12 What is gained and what is lost

The advantages of log-linearization are substantial.

- (i) It produces analytically transparent equilibrium systems.
- (ii) It makes impulse responses easy to compute and interpret.
- (iii) It isolates the main local transmission mechanisms around steady state.
- (iv) It often allows one to compare different models using the same reduced-form structure.

But the costs are also real.

- (i) It is a local approximation, valid only near the chosen steady state.
- (ii) It suppresses nonlinear amplification and asymmetry.
- (iii) It removes many precautionary and risk-related terms at first order.
- (iv) It hides some distribution-dependent mechanisms that matter in heterogeneous-agent models.
- (v) It can be misleading for large shocks, occasionally binding constraints, or welfare analysis.

5.13 Why this matters for the models in this manual

This manual uses log-linearization for two reasons. First, it provides the cleanest way to see how the main equilibrium objects are connected. Second, much of the classic New Keynesian literature is written in log-linear form, so students need to become fluent in it.

At the same time, one should not confuse the linearized system with the full model. In RANK and some TANK environments, the linearized representation often captures much of the relevant intuition. In THANK and HANK, however, one must be more cautious. Some of the economically interesting mechanisms are precisely those that become weak or invisible under first-order approximation.

For that reason, log-linearization should be seen as a tool of understanding, not as a complete substitute for the nonlinear model.

6 Roadmap: what changes from RANK to TANK to THANK to HANK

Before turning to equations, it is useful to describe the progression in plain language.

In a representative-agent New Keynesian model, the household sector is summarized by one Euler equation and one labor-supply condition. Monetary policy works by changing the real interest rate, which alters the intertemporal price of consumption. Because there is only one household, there is no distinction between aggregate and distributional responses.

In a two-agent New Keynesian model, the household side is split into a group that smooths consumption intertemporally and a group that does not. Monetary policy still affects aggregate demand through the real rate, but now part of the economy reacts mainly to current disposable income rather than to intertemporal substitution.

In a tractable HANK model, which we call THANK, households still belong to a small number of states, but the model allows one to capture risk, precautionary saving, self-insurance, cyclical inequality, and realistic intertemporal MPCs in an analytically transparent way.

In a full HANK model, households differ in asset holdings and idiosyncratic income, face incomplete markets, and are summarized by policy functions and a cross-sectional distribution. Monetary and fiscal policy then operate through several interacting channels at once.

Table 1: Conceptual roadmap from RANK to TANK to THANK to HANK

Model	Who smooths consumption?	What heterogeneity matters?	What extra mechanisms become active?
RANK	Everyone, because there is one representative household	None	Pure intertemporal substitution and standard NK general equilibrium
TANK	Only Ricardian households	A discrete split between high-MPC and low-MPC households	Heterogeneous disposable-income responses and basic MPC heterogeneity
THANK	Some states smooth, others are constrained; transitions matter	Few states, but with risk and self-insurance logic	Cyclical inequality, risk, precautionary saving, richer iMPCs, stronger policy incidence
HANK	Heterogeneously across the whole distribution	Full distribution of assets and idiosyncratic income	Precautionary saving, endogenous wealth distribution, redistribution, state-dependent responses, full incomplete-markets transmission

7 Common New Keynesian structure

All four environments in this manual share the same basic nominal and production side. Time is discrete and indexed by $t = 0, 1, 2, \dots$. The economy is closed. There is a competitive final-good producer, a continuum of monopolistically competitive intermediate firms, sticky prices, a monetary authority that follows a Taylor rule, and a fiscal authority.

7.1 Overview of the common structure

The common structure can be summarized as follows.

- (i) Final-good firms aggregate differentiated varieties into a final good.
- (ii) Intermediate firms produce with labor under monopolistic competition.
- (iii) Sticky prices create a Phillips curve.
- (iv) Monetary policy affects the nominal interest rate through a rule.

- (v) The real interest rate connects monetary policy to household decisions.
- (vi) The government supplies safe assets and redistributes profits and taxes.

The key pedagogical strategy is to keep this side of the model fixed as much as possible and vary the household side.

The firm side is the nominal core that remains common across the models studied in this manual. This is deliberate. If one changed both the household block and the firm block at the same time, it would become much harder to see which mechanisms come from household heterogeneity and which come from the pricing side. For that reason, we develop the production and pricing structure once, in detail, and then reuse it throughout the RANK, TANK, THANK, and HANK environments.

The purpose of the firm block is threefold. First, it generates a well-defined demand schedule for each differentiated producer. Second, it links wages and productivity to real marginal cost. Third, under sticky pricing, it delivers the New Keynesian Phillips curve that governs inflation dynamics. Since these objects will reappear repeatedly in later sections, it is worth deriving them slowly and explicitly.

8 Firm-side foundations

This section develops the firm side in detail. The goal is not only to state the pricing equations used later in the model, but also to explain where they come from and why they matter. The logic runs in three steps.

First, a perfectly competitive final-good sector aggregates a continuum of differentiated intermediate goods into a single composite good. This aggregation problem delivers a demand schedule for each variety and an associated aggregate price index. Second, intermediate-good firms produce their own differentiated varieties using labor. Because each firm supplies a distinct variety, it faces a downward-sloping demand curve and therefore has market power. Third, depending on whether prices are flexible or sticky, the firm either chooses a static profit-maximizing price or solves a dynamic price-setting problem of the Calvo type.

Throughout, $\varepsilon_p > 1$ denotes the elasticity of substitution across differentiated goods. A larger ε_p means that goods are closer substitutes, demand is more price-elastic, and firms have less monopoly power.

8.1 Final-good firms

A representative competitive final-good firm combines a continuum of differentiated intermediate varieties $Y_t(i)$, indexed by $i \in [0, 1]$, into a single final good Y_t according to the CES aggregator

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} di \right)^{\frac{\varepsilon_p}{\varepsilon_p - 1}}. \quad (34)$$

This specification is standard in New Keynesian models. It is chosen because it implies a constant elasticity of substitution across varieties and therefore generates simple isoelastic demand functions for each intermediate good.

The final-good firm is perfectly competitive. Taking all prices $\{P_t(i)\}_{i \in [0,1]}$ and the aggregate output target Y_t as given, it chooses the bundle $\{Y_t(i)\}_{i \in [0,1]}$ that minimizes the total cost of producing one unit of the final good:

$$\min_{\{Y_t(i)\}_{i \in [0,1]}} \int_0^1 P_t(i) Y_t(i) \, di \quad (35)$$

subject to (34).

The associated Lagrangian is

$$\mathcal{L}^F = \int_0^1 P_t(i) Y_t(i) \, di + \Lambda_t \left[Y_t - \left(\int_0^1 Y_t(i)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \, di \right)^{\frac{\varepsilon_p}{\varepsilon_p - 1}} \right]. \quad (36)$$

Here Λ_t is the multiplier on the production constraint. It will turn out to be the nominal marginal cost of producing one additional unit of the final good, which under perfect competition equals the final-good price index.

Lemma 8.1. *The optimal demand for variety i is*

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t, \quad (37)$$

where the aggregate price index is

$$P_t = \left(\int_0^1 P_t(i)^{1-\varepsilon_p} \, di \right)^{\frac{1}{1-\varepsilon_p}}. \quad (38)$$

Proof. Differentiate the Lagrangian with respect to $Y_t(i)$. Using the chain rule,

$$\begin{aligned} \frac{\partial \mathcal{L}^F}{\partial Y_t(i)} &= P_t(i) - \Lambda_t \frac{\varepsilon_p}{\varepsilon_p - 1} \left(\int_0^1 Y_t(j)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \, dj \right)^{\frac{1}{\varepsilon_p - 1}} \cdot \frac{\varepsilon_p - 1}{\varepsilon_p} Y_t(i)^{-1/\varepsilon_p} \\ &= P_t(i) - \Lambda_t \left(\int_0^1 Y_t(j)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \, dj \right)^{\frac{1}{\varepsilon_p - 1}} Y_t(i)^{-1/\varepsilon_p}. \end{aligned} \quad (39)$$

The first-order condition is therefore

$$P_t(i) = \Lambda_t \left(\int_0^1 Y_t(j)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \, dj \right)^{\frac{1}{\varepsilon_p - 1}} Y_t(i)^{-1/\varepsilon_p}. \quad (40)$$

From the aggregator (34),

$$Y_t^{\frac{\varepsilon_p - 1}{\varepsilon_p}} = \int_0^1 Y_t(j)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \, dj,$$

so

$$\left(\int_0^1 Y_t(j)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \, dj \right)^{\frac{1}{\varepsilon_p - 1}} = Y_t^{1/\varepsilon_p}.$$

Substituting this into (40) yields

$$P_t(i) = \Lambda_t Y_t^{1/\varepsilon_p} Y_t(i)^{-1/\varepsilon_p}. \quad (41)$$

Rearranging,

$$Y_t(i) = \left(\frac{\Lambda_t}{P_t(i)} \right)^{\varepsilon_p} Y_t. \quad (42)$$

To determine Λ_t , substitute (42) into the CES aggregator:

$$\begin{aligned} Y_t &= \left(\int_0^1 \left[\left(\frac{\Lambda_t}{P_t(i)} \right)^{\varepsilon_p} Y_t \right]^{\frac{\varepsilon_p-1}{\varepsilon_p}} di \right)^{\frac{\varepsilon_p}{\varepsilon_p-1}} \\ &= \left(\int_0^1 \left(\frac{\Lambda_t}{P_t(i)} \right)^{\varepsilon_p-1} Y_t^{\frac{\varepsilon_p-1}{\varepsilon_p}} di \right)^{\frac{\varepsilon_p}{\varepsilon_p-1}}. \end{aligned} \quad (43)$$

Factor out the terms that do not depend on i :

$$\begin{aligned} Y_t &= \left[\Lambda_t^{\varepsilon_p-1} Y_t^{\frac{\varepsilon_p-1}{\varepsilon_p}} \int_0^1 P_t(i)^{1-\varepsilon_p} di \right]^{\frac{\varepsilon_p}{\varepsilon_p-1}} \\ &= \Lambda_t^{\varepsilon_p} Y_t \left(\int_0^1 P_t(i)^{1-\varepsilon_p} di \right)^{\frac{\varepsilon_p}{\varepsilon_p-1}}. \end{aligned} \quad (44)$$

Cancelling Y_t from both sides,

$$1 = \Lambda_t^{\varepsilon_p} \left(\int_0^1 P_t(i)^{1-\varepsilon_p} di \right)^{\frac{\varepsilon_p}{\varepsilon_p-1}}, \quad (45)$$

which implies

$$\Lambda_t = \left(\int_0^1 P_t(i)^{1-\varepsilon_p} di \right)^{\frac{1}{1-\varepsilon_p}} \equiv P_t. \quad (46)$$

Substituting $\Lambda_t = P_t$ into (42) gives

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t, \quad (47)$$

which is the desired demand function. $\square$

The expression for P_t in (38) is the minimum unit expenditure required to purchase one unit of the final good. Because the final-good sector is competitive, zero profits imply that this unit expenditure is also the market price of the final good.

8.2 Interpretation

The final-good block is mainly a device for generating demand schedules for intermediate producers. Its economic role is easy to summarize.

Because each intermediate firm sells a differentiated product, it does not face perfectly elastic demand. Instead, it faces the isoelastic demand curve (37). The elasticity of that demand curve is exactly ε_p . This is what gives firms monopoly power and makes pricing decisions meaningful. If goods were homogeneous, each firm would be a price taker and nominal rigidities at the firm level would have no bite.

The parameter ε_p summarizes the intensity of product-market competition. When ε_p is high, demand is very sensitive to relative prices, so firms want to keep their prices close to competitors' prices. When ε_p is low, each firm has more local market power and can sustain a larger markup over marginal cost.

8.3 Intermediate-good firms

Intermediate firm i produces output linearly with labor:

$$Y_t(i) = A_t N_t(i), \quad (48)$$

where A_t is aggregate labor productivity and $N_t(i)$ is labor employed by firm i .

The linear technology keeps the analysis transparent. Because one extra unit of labor produces exactly A_t extra units of output, the marginal product of labor is constant and equal to A_t .

Nominal profits are

$$\Pi_t(i) = P_t(i) Y_t(i) - W_t N_t(i), \quad (49)$$

where W_t is the nominal wage. Using (48), labor demand can be written as

$$N_t(i) = \frac{Y_t(i)}{A_t}, \quad (50)$$

so profits become

$$\Pi_t(i) = \left[P_t(i) - \frac{W_t}{A_t} \right] Y_t(i). \quad (51)$$

This expression shows immediately that nominal marginal cost is W_t/A_t : producing one extra unit of output requires $1/A_t$ extra units of labor, each paid at wage W_t .

Dividing by the aggregate price level P_t , real marginal cost is

$$\text{mc}_t = \frac{W_t/P_t}{A_t} = \frac{w_t}{A_t}, \quad w_t = \frac{W_t}{P_t}. \quad (52)$$

Real marginal cost is a central variable in New Keynesian pricing equations. Intuitively, it measures the real resource cost of producing an additional unit of output. Inflation will be linked to marginal cost because firms want higher prices whenever producing is relatively expensive.

8.4 Flexible-price pricing

Suppose first that prices are fully flexible, so every firm can reoptimize its price each period. Firm i chooses $P_t(i)$ to maximize current nominal profits, taking aggregate demand Y_t , the price level P_t , the nominal wage W_t , and productivity A_t as given. Using the demand function (37), profits are

$$\Pi_t(i) = P_t(i) \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t - W_t \frac{1}{A_t} \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t. \quad (53)$$

Equivalently,

$$\Pi_t(i) = \left[P_t(i) - \frac{W_t}{A_t} \right] \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t. \quad (54)$$

Proposition 8.1. *Under flexible prices, the optimal price is a constant markup over nominal marginal cost:*

$$P_t(i) = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{W_t}{A_t}. \quad (55)$$

Hence real marginal cost under flexible prices is

$$\text{mc}_t = \frac{\varepsilon_p - 1}{\varepsilon_p} \frac{P_t(i)}{P_t}. \quad (56)$$

In a symmetric equilibrium, where all firms charge the same price and therefore $P_t(i) = P_t$, this reduces to

$$\text{mc}_t = \frac{\varepsilon_p - 1}{\varepsilon_p}. \quad (57)$$

Proof. Let

$$MC_t^n \equiv \frac{W_t}{A_t}$$

denote nominal marginal cost. Then

$$\Pi_t(i) = [P_t(i) - MC_t^n] \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t. \quad (58)$$

Since P_t and Y_t are taken as given by an individual firm, maximize with respect to $P_t(i)$. Write

$$D_t(i) \equiv \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t.$$

Then

$$\Pi_t(i) = [P_t(i) - MC_t^n] D_t(i).$$

By the product rule,

$$\frac{\partial \Pi_t(i)}{\partial P_t(i)} = D_t(i) + [P_t(i) - MC_t^n] \frac{\partial D_t(i)}{\partial P_t(i)}. \quad (59)$$

Now

$$\frac{\partial D_t(i)}{\partial P_t(i)} = -\varepsilon_p \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} \frac{Y_t}{P_t(i)} = -\varepsilon_p \frac{D_t(i)}{P_t(i)}. \quad (60)$$

Substituting (60) into (59) and setting the result equal to zero:

$$D_t(i) - \varepsilon_p [P_t(i) - MC_t^n] \frac{D_t(i)}{P_t(i)} = 0. \quad (61)$$

Divide by $D_t(i) > 0$:

$$1 - \varepsilon_p \frac{P_t(i) - MC_t^n}{P_t(i)} = 0. \quad (62)$$

Rearranging,

$$P_t(i) = \varepsilon_p [P_t(i) - MC_t^n], \quad (63)$$

so

$$(\varepsilon_p - 1)P_t(i) = \varepsilon_p MC_t^n. \quad (64)$$

Therefore,

$$P_t(i) = \frac{\varepsilon_p}{\varepsilon_p - 1} MC_t^n = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{W_t}{A_t}. \quad (65)$$

Finally, divide both sides by P_t :

$$\frac{P_t(i)}{P_t} = \frac{\varepsilon_p}{\varepsilon_p - 1} mc_t. \quad (66)$$

In a symmetric equilibrium, $P_t(i) = P_t$, so (57) follows immediately. $\square$

This markup formula is one of the core objects in the model. The desired flexible price is higher when nominal marginal cost is higher, and the gap between price and marginal cost is governed by the markup factor $\varepsilon_p/(\varepsilon_p - 1)$. The corresponding gross markup shrinks as goods become closer substitutes.

8.5 Sticky-price firms

We now move from flexible prices to nominal rigidity. Under Calvo pricing, each intermediate firm receives an opportunity to reset its price in period t with probability $1 - \theta_p$. With probability θ_p , it cannot reoptimize and must keep its existing price in place.

Thus, a firm that is allowed to reoptimize at date t chooses a reset price P_t^* knowing that, if it is not selected again in future periods, that same price may remain in effect for several periods. The price-setting problem is therefore forward-looking.

A firm resetting in period t solves

$$\max_{P_t^*} \mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} \left[P_t^* Y_{t+s|t} - W_{t+s} N_{t+s|t} \right], \quad (67)$$

subject to

$$Y_{t+s|t} = \left(\frac{P_t^*}{P_{t+s}} \right)^{-\varepsilon_p} Y_{t+s}, \quad (68)$$

and

$$Y_{t+s|t} = A_{t+s} N_{t+s|t}. \quad (69)$$

Several terms deserve comment.

First, $Y_{t+s|t}$ denotes demand in period $t + s$ for a firm that last reset its price in period t . Second, $(\theta_p)^s$ is the probability that the price chosen in period t is still in place in period $t + s$. Third, $\mathcal{M}_{t,t+s}$ is the stochastic discount factor used by households to value nominal profit streams. Since firms are owned by households, future profits are discounted using the household's intertemporal marginal rate of substitution.

Combining (68) and (69), nominal profits in period $t + s$ for a firm that reset at date t can be written as

$$\left[P_t^* - \frac{W_{t+s}}{A_{t+s}} \right] \left(\frac{P_t^*}{P_{t+s}} \right)^{-\varepsilon_p} Y_{t+s}. \quad (70)$$

Using real marginal cost $mc_{t+s} = W_{t+s}/(P_{t+s}A_{t+s})$, this becomes

$$[P_t^* - P_{t+s}mc_{t+s}] \left(\frac{P_t^*}{P_{t+s}} \right)^{-\varepsilon_p} Y_{t+s}. \quad (71)$$

Therefore the reset-price problem can be written as

$$\max_{P_t^*} \mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} [P_t^* - P_{t+s}mc_{t+s}] \left(\frac{P_t^*}{P_{t+s}} \right)^{-\varepsilon_p} Y_{t+s}. \quad (72)$$

8.6 Reset-price condition

Because the chosen price P_t^* may stay fixed for several periods, the firm trades off current and future considerations. Setting a higher price raises revenue per unit sold, but lowers demand in every future period in which the price remains in place. The optimal reset price balances these forces.

Proposition 8.2. *The first-order condition for the optimal reset price is*

$$\mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} Y_{t+s} P_{t+s}^{\varepsilon_p} (P_t^*)^{-\varepsilon_p} [(1 - \varepsilon_p) P_t^* + \varepsilon_p P_{t+s}mc_{t+s}] = 0. \quad (73)$$

Equivalently, the optimal reset price can be written as

$$P_t^* = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{\mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} P_{t+s}^{\varepsilon_p} Y_{t+s} mc_{t+s}}{\mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} P_{t+s}^{\varepsilon_p - 1} Y_{t+s}}. \quad (74)$$

Proof. Start from (72). For fixed s , define

$$f_s(P_t^*) = [P_t^* - P_{t+s}mc_{t+s}] \left(\frac{P_t^*}{P_{t+s}} \right)^{-\varepsilon_p} Y_{t+s}. \quad (75)$$

Since

$$\left(\frac{P_t^*}{P_{t+s}} \right)^{-\varepsilon_p} = (P_t^*)^{-\varepsilon_p} P_{t+s}^{\varepsilon_p},$$

we can write

$$f_s(P_t^*) = [P_t^* - P_{t+s}mc_{t+s}] (P_t^*)^{-\varepsilon_p} P_{t+s}^{\varepsilon_p} Y_{t+s}. \quad (76)$$

Differentiate with respect to P_t^* :

$$\begin{aligned} \frac{\partial f_s(P_t^*)}{\partial P_t^*} &= P_{t+s}^{\varepsilon_p} Y_{t+s} \frac{\partial}{\partial P_t^*} \{ [P_t^* - P_{t+s}mc_{t+s}] (P_t^*)^{-\varepsilon_p} \} \\ &= P_{t+s}^{\varepsilon_p} Y_{t+s} \left[(P_t^*)^{-\varepsilon_p} - \varepsilon_p (P_t^* - P_{t+s}mc_{t+s}) (P_t^*)^{-\varepsilon_p - 1} \right]. \end{aligned} \quad (77)$$

Factor out $(P_t^*)^{-\varepsilon_p-1}$:

$$\begin{aligned}\frac{\partial f_s(P_t^*)}{\partial P_t^*} &= P_{t+s}^{\varepsilon_p} Y_{t+s} (P_t^*)^{-\varepsilon_p-1} [P_t^* - \varepsilon_p (P_t^* - P_{t+s} \text{mc}_{t+s})] \\ &= P_{t+s}^{\varepsilon_p} Y_{t+s} (P_t^*)^{-\varepsilon_p-1} [(1 - \varepsilon_p) P_t^* + \varepsilon_p P_{t+s} \text{mc}_{t+s}].\end{aligned}\quad (78)$$

Multiplying by P_t^* , which is strictly positive, gives an equivalent expression:

$$\frac{\partial f_s(P_t^*)}{\partial P_t^*} = P_{t+s}^{\varepsilon_p} Y_{t+s} (P_t^*)^{-\varepsilon_p} [(1 - \varepsilon_p) P_t^* + \varepsilon_p P_{t+s} \text{mc}_{t+s}]. \quad (79)$$

Substituting into the first-order condition for (72) yields (73).

To obtain the closed-form expression for P_t^* , distribute the summation across the two terms inside brackets:

$$\begin{aligned}0 &= (1 - \varepsilon_p) P_t^* \mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} Y_{t+s} P_{t+s}^{\varepsilon_p} (P_t^*)^{-\varepsilon_p} \\ &\quad + \varepsilon_p \mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} Y_{t+s} P_{t+s}^{\varepsilon_p+1} (P_t^*)^{-\varepsilon_p} \text{mc}_{t+s}.\end{aligned}\quad (80)$$

Factor out the common term $(P_t^*)^{-\varepsilon_p}$ and divide through by it:

$$(\varepsilon_p - 1) P_t^* \mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} Y_{t+s} P_{t+s}^{\varepsilon_p} = \varepsilon_p \mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} Y_{t+s} P_{t+s}^{\varepsilon_p+1} \text{mc}_{t+s}. \quad (81)$$

Finally divide both sides by

$$(\varepsilon_p - 1) \mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} Y_{t+s} P_{t+s}^{\varepsilon_p},$$

and equivalently factor one P_{t+s} out of numerator and denominator to write the result as

$$P_t^* = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{\mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} P_{t+s}^{\varepsilon_p} Y_{t+s} \text{mc}_{t+s}}{\mathbb{E}_t \sum_{s=0}^{\infty} (\theta_p)^s \mathcal{M}_{t,t+s} P_{t+s}^{\varepsilon_p-1} Y_{t+s}}, \quad (82)$$

which is (74). □

Equation (74) has a very intuitive structure. It says that the reset price is a markup over a weighted average of expected future nominal marginal costs. The weights are larger when future states are more likely to occur, when households value profits in those states more highly, and when the firm's future demand is larger.

The dynamic nature of the problem is essential. A firm does not choose its price only for the present period. It chooses a price that may remain fixed while aggregate prices and marginal costs continue to evolve. This is why expected future marginal costs enter the reset-price formula and, ultimately, why inflation is forward-looking.

8.7 Calvo aggregation and the Phillips curve

Let P_t^* denote the common reset price chosen by all firms that are allowed to reoptimize at date t . Since a fraction $1 - \theta_p$ of firms reset and a fraction θ_p keep their previous prices, the aggregate price index satisfies

$$P_t^{1-\varepsilon_p} = (1 - \theta_p)(P_t^*)^{1-\varepsilon_p} + \theta_p P_{t-1}^{1-\varepsilon_p}. \quad (83)$$

This equation is simply the CES price index applied to the cross-section of firms. A measure $1 - \theta_p$ of firms charge the newly chosen price P_t^* , while the remaining measure θ_p continue charging the price they had in the previous period.

The exact nonlinear system consists of (74) together with (83). To obtain a tractable inflation equation, one log-linearizes these conditions around a zero-inflation symmetric steady state. The resulting New Keynesian Phillips curve is

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa \hat{m}c_t, \quad (84)$$

where $\pi_t \equiv \log P_t - \log P_{t-1}$ and $\hat{m}c_t$ is log-deviation of real marginal cost from steady state.

In the basic Calvo model without indexation,

$$\kappa = \frac{(1 - \theta_p)(1 - \beta\theta_p)}{\theta_p}. \quad (85)$$

If one rewrites inflation in terms of an output gap rather than marginal cost, an additional proportionality factor appears because marginal cost itself must be linked to output using households' labor-supply condition and firms' technology.

Equation (84) is central to New Keynesian analysis. It states that current inflation depends on two forces. The first is expected future inflation, reflecting the forward-looking nature of reset-price decisions. The second is real marginal cost, reflecting the incentive to raise prices when production is costly relative to the final-good price.

8.8 Price dispersion

An important implication of staggered pricing is that identical firms generally charge different prices. Once firms charge different prices, aggregate production is no longer a simple linear function of aggregate labor. The reason is that demand is inefficiently spread across firms: low-price firms sell too much, high-price firms sell too little, and labor is not allocated evenly.

A convenient summary statistic is the price-dispersion term

$$\Delta_t = \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} di. \quad (86)$$

Using the firm-level production function $Y_t(i) = A_t N_t(i)$ and the demand schedule (37), aggregate

labor satisfies

$$\begin{aligned} N_t &= \int_0^1 N_t(i) \, di = \int_0^1 \frac{Y_t(i)}{A_t} \, di \\ &= \frac{1}{A_t} \int_0^1 \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon_p} Y_t \, di = \frac{Y_t}{A_t} \Delta_t. \end{aligned} \quad (87)$$

Thus aggregate output can be written as

$$Y_t = \frac{A_t N_t}{\Delta_t}. \quad (88)$$

When all firms charge the same price, $\Delta_t = 1$ and the usual aggregate production relation $Y_t = A_t N_t$ holds. But when prices differ across firms, $\Delta_t > 1$, so more labor is needed to produce a given level of final output. Price dispersion therefore acts like a wedge or inefficiency term.

This is one of the key welfare costs of inflation in sticky-price models. Inflation is not costly only because the average price level changes. It is costly because staggered adjustment creates relative-price distortions across otherwise identical firms, and those distortions reduce aggregate productive efficiency.

In a zero-inflation steady state all firms charge the same price, so $\Delta = 1$. Around that steady state, dispersion is a second-order object, which is why it disappears from first-order log-linearized versions of the model but matters for welfare and for nonlinear dynamics.

9 RANK: representative-agent New Keynesian model

We begin with the representative-agent benchmark. This model is the natural starting point for two reasons. First, it is the cleanest environment in which to see the interaction between intertemporal household choice, sticky-price inflation dynamics, and monetary policy. Second, all later models in this manual can be read as modifications of this benchmark household block while keeping much of the nominal side unchanged.

The representative-agent New Keynesian model, or RANK, is not interesting because it is realistic in a literal cross-sectional sense. It is interesting because it isolates the aggregate logic of the modern New Keynesian framework as transparently as possible. There is one household, one Euler equation, one labor-supply condition, one aggregate consumption level, and no internal redistribution across households. Monetary policy works mainly through the real interest rate, which affects intertemporal substitution in consumption and, through aggregate demand, firms' pricing decisions.

9.1 Household problem

The representative household chooses sequences $\{C_t, B_{t+1}, N_t\}_{t \geq 0}$ to maximize expected discounted utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma} - 1}{1-\sigma} - \chi \frac{N_t^{1+\phi_n}}{1+\phi_n} \right], \quad (89)$$

subject to the period-by-period budget constraint

$$C_t + B_{t+1} = R_t B_t + w_t N_t + \Pi_t / P_t - \tau_t. \quad (90)$$

Several objects deserve interpretation.

Consumption is denoted by C_t , labor supply by N_t , and bond holdings chosen at date t for payoff at date $t + 1$ by B_{t+1} . The parameter $\beta \in (0, 1)$ is the discount factor. The coefficient $\sigma > 0$ governs the curvature of utility over consumption and therefore the strength of intertemporal substitution. The parameter $\phi_n > 0$ governs the curvature of the disutility of labor, and $\chi > 0$ scales the disutility of working.

On the income side, the household receives gross payoff $R_t B_t$ from bond holdings carried into period t , labor income $w_t N_t$, real firm profits Π_t / P_t , and pays lump-sum taxes τ_t . Here $w_t = W_t / P_t$ is the real wage. The bond is a one-period safe asset in zero net supply unless the fiscal block specifies otherwise.

Because the economy contains only one household, this representative household owns firms, receives all profits, and ultimately holds the aggregate safe asset position of the private sector.

9.2 Lagrangian

The Lagrangian for the household problem is

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \chi \frac{N_t^{1+\phi_n}}{1+\phi_n} + \lambda_t (R_t B_t + w_t N_t + \Pi_t / P_t - \tau_t - C_t - B_{t+1}) \right\}, \quad (91)$$

where λ_t is the multiplier on the budget constraint in period t .

Economically, λ_t is the marginal value of one additional unit of real wealth at date t . Since an extra unit of wealth can be converted into an extra unit of consumption, this multiplier will turn out to equal the marginal utility of consumption.

9.3 First-order conditions

Differentiating with respect to C_t , N_t , and B_{t+1} yields the first-order conditions

$$C_t^{-\sigma} = \lambda_t, \quad (92)$$

$$\chi N_t^{\phi_n} = \lambda_t w_t, \quad (93)$$

$$\lambda_t = \beta \mathbb{E}_t [R_{t+1} \lambda_{t+1}]. \quad (94)$$

The bond first-order condition is written one period ahead to make the timing transparent. Choosing one extra unit of B_{t+1} costs one unit of resources today and delivers gross payoff R_{t+1} tomorrow, valued at tomorrow's marginal utility of wealth.

Proposition 9.1. *The representative household satisfies the Euler equation*

$$C_t^{-\sigma} = \beta \mathbb{E}_t [R_{t+1} C_{t+1}^{-\sigma}] \quad (95)$$

and the intratemporal labor-supply condition

$$w_t = \chi N_t^{\phi_n} C_t^\sigma. \quad (96)$$

Proof. From (92), the multiplier equals marginal utility of consumption:

$$\lambda_t = C_t^{-\sigma}.$$

Substituting this into the bond first-order condition (94) gives

$$C_t^{-\sigma} = \beta \mathbb{E}_t [R_{t+1} C_{t+1}^{-\sigma}],$$

which is (95). Substituting the same expression for λ_t into the labor first-order condition (93) yields

$$\chi N_t^{\phi_n} = C_t^{-\sigma} w_t,$$

which can be rearranged as

$$w_t = \chi N_t^{\phi_n} C_t^\sigma.$$

□

9.4 Interpretation

The Euler equation (95) is the core behavioral equation of the representative household. It states that the household must be indifferent at the margin between consuming one more unit today and saving that unit in the safe asset to consume tomorrow. If the expected real return on saving rises, the household has an incentive to postpone consumption, so current consumption falls relative to future consumption.

The labor-supply condition (96) is an intratemporal optimality condition. It equates the real wage to the marginal rate of substitution between consumption and labor. The household supplies labor up to the point at which the utility gain from the extra consumption financed by working equals the utility cost of supplying the extra unit of labor.

These two conditions summarize the household side of RANK. This is one of the main reasons the model is analytically attractive: the entire private household block collapses to one intertemporal condition and one intratemporal condition.

9.5 A note on the real return

To connect the household problem to monetary policy, it is useful to distinguish carefully between nominal and real returns. If the gross nominal interest rate between t and $t+1$ is denoted by $1+i_t$, then the ex post gross real return is

$$R_{t+1} = \frac{1+i_t}{1+\pi_{t+1}}, \quad (97)$$

where $\pi_{t+1} \equiv P_{t+1}/P_t - 1$ is gross inflation expressed in net terms when convenient.

Substituting (97) into the Euler equation shows immediately why monetary policy matters in

the New Keynesian model. The central bank moves the nominal interest rate, expected inflation determines how much of that nominal return translates into a real return, and the resulting real return affects intertemporal consumption choices.

9.6 Goods-market clearing and aggregation

Because there is one representative household and one final good, aggregation is especially simple. In the absence of capital accumulation and abstracting for the moment from government purchases and adjustment costs, aggregate resource feasibility implies

$$Y_t = C_t. \quad (98)$$

If later sections add government spending or other resource uses, this condition must be modified accordingly. But in the baseline benchmark, output equals aggregate consumption.

Labor market aggregation is also straightforward:

$$N_t = \int_0^1 N_t(i) di. \quad (99)$$

Using the firm-side block developed earlier, real marginal cost is linked to the real wage and productivity by

$$mc_t = \frac{w_t}{A_t}. \quad (100)$$

Combining this with household labor supply gives

$$mc_t = \frac{\chi N_t^{\phi_n} C_t^\sigma}{A_t}. \quad (101)$$

This relation is important because it connects the household side to firms' pricing decisions. In RANK, fluctuations in aggregate consumption and labor move marginal cost, and marginal cost in turn moves inflation through the Phillips curve.

9.7 Equilibrium in RANK

A competitive equilibrium in the representative-agent New Keynesian economy is a sequence

$$\{Y_t, C_t, N_t, w_t, mc_t, \pi_t, i_t, R_{t+1}, B_{t+1}\}_{t \geq 0} \quad (102)$$

such that:

- (i) given prices, profits, taxes, and returns, the representative household solves its optimization problem;
- (ii) final-good firms and intermediate-good firms satisfy the production and pricing conditions derived in Sections 7 and 8;
- (iii) monetary policy sets the nominal interest rate according to a Taylor rule;
- (iv) the goods market clears;

(v) the bond market clears.

Because there is only one household, bond-market clearing is especially simple. If government debt is zero in net supply, then in equilibrium

$$B_{t+1} = 0 \quad (103)$$

for all t . If government debt is positive, the representative household holds the entire supply.

The important point is that in RANK there is no internal distribution to keep track of. Aggregate behavior and individual behavior coincide. This is exactly what later heterogeneous-agent models will break.

9.8 From the Euler equation to the dynamic IS relation

The nonlinear Euler equation is

$$C_t^{-\sigma} = \beta \mathbb{E}_t \left[\frac{1 + i_t}{1 + \pi_{t+1}} C_{t+1}^{-\sigma} \right]. \quad (104)$$

To obtain the standard New Keynesian IS relation, one log-linearizes this equation around a zero-inflation steady state with constant consumption. The result is

$$\hat{c}_t = \mathbb{E}_t[\hat{c}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t[\pi_{t+1}] \right). \quad (105)$$

Under the baseline resource constraint $Y_t = C_t$, output and consumption move one for one, so

$$\hat{y}_t = \hat{c}_t.$$

Hence the dynamic IS equation can be written as

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t[\pi_{t+1}] \right). \quad (106)$$

In many presentations, this equation is expressed in output-gap form:

$$\hat{x}_t = \mathbb{E}_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t[\pi_{t+1}] - r_t^{\text{nat}} \right), \quad (107)$$

where $\hat{x}_t$ is the output gap and r_t^{nat} is the natural real rate, that is, the real interest rate consistent with flexible prices and zero output gap. This formulation is often preferable because it separates demand fluctuations from movements in the flexible-price benchmark.

9.9 Linearized RANK system

Around a zero-inflation steady state, the benchmark RANK model can be summarized by three equations.

The first is the dynamic IS relation:

$$\hat{x}_t = \mathbb{E}_t[\hat{x}_{t+1}] - \frac{1}{\sigma} \left(\hat{i}_t - \mathbb{E}_t[\pi_{t+1}] - r_t^{\text{nat}} \right). \quad (108)$$

The second is the New Keynesian Phillips curve:

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa \hat{x}_t. \quad (109)$$

The third is the Taylor rule:

$$\hat{i}_t = \phi_\pi \pi_t + \phi_y \hat{x}_t + \nu_t. \quad (110)$$

Taken together, these three equations form the canonical log-linear New Keynesian system. The IS curve comes from the representative household's Euler equation. The Phillips curve comes from firms' Calvo pricing problem. The Taylor rule comes from monetary policy. The equilibrium dynamics of output, inflation, and the nominal interest rate are determined jointly by these three blocks.

9.10 Interpretation of the linearized system

This compact three-equation system is one of the reasons RANK became the benchmark model for modern monetary macroeconomics.

A contractionary monetary shock $\nu_t > 0$ raises the nominal interest rate. For given expected inflation, this raises the real interest rate. The representative household then wants to postpone consumption, which lowers current aggregate demand and opens a negative output gap. Lower demand reduces labor demand and marginal cost. Through the Phillips curve, lower marginal cost reduces inflation. The fall in inflation partly offsets the rise in the nominal rate by lowering expected inflation, but unless that offset is complete, the real rate still rises on impact and demand contracts.

The logic is powerful but narrow. Monetary transmission in RANK works mainly through the intertemporal substitution channel. There is no internal redistribution across households, no high-MPC and low-MPC groups, no borrowing-constraint heterogeneity, and no distinction between aggregate responses and cross-sectional responses. Those missing mechanisms are exactly what motivate the move to TANK, THANK, and HANK.

9.11 Figures for RANK

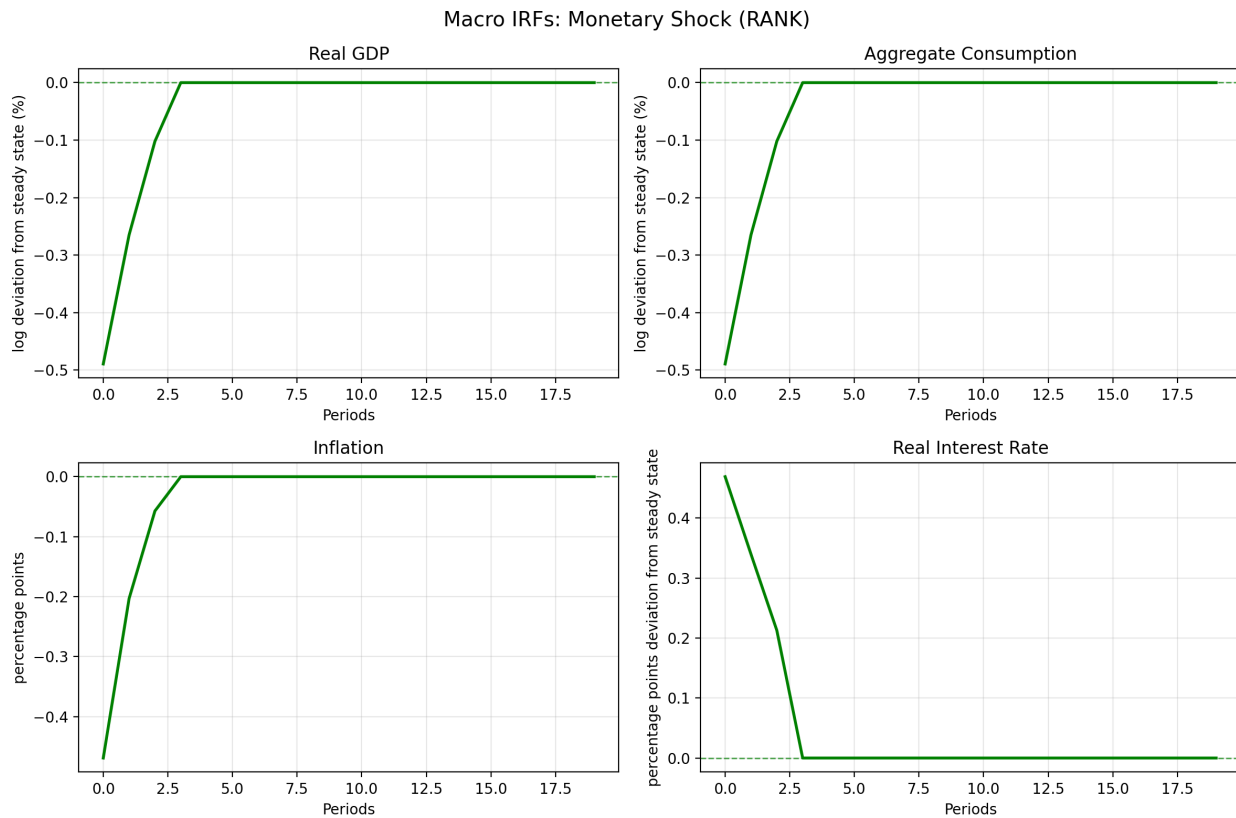


Figure 1: RANK macro impulse responses to a contractionary monetary shock.

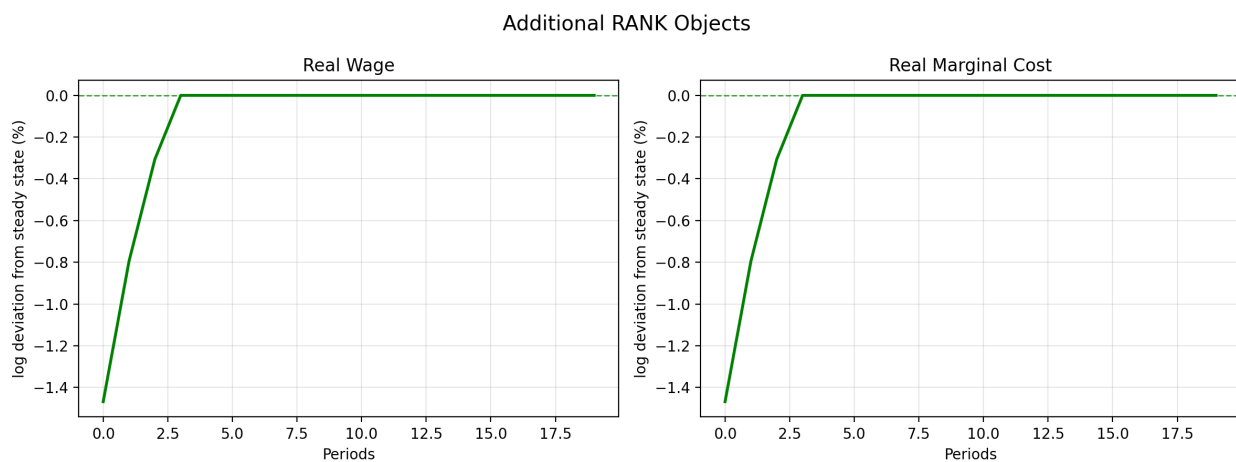


Figure 2: Additional RANK transition objects.

9.12 Reading the figures

The impulse responses summarize the transmission logic of the representative-agent benchmark. A contractionary monetary shock raises the policy rate, increases the real interest rate, induces

the household to postpone consumption, lowers current demand, reduces output and labor, pushes down real marginal cost, and lowers inflation.

The additional transition objects are useful because they show variables that are often hidden once the model is reduced to the textbook three-equation system. In particular, they help the reader see how wages, marginal cost, and the real return move during the adjustment.

10 TANK: two-agent New Keynesian model

The next step introduces the simplest possible form of household heterogeneity. A fraction $\lambda \in (0, 1)$ of households is hand-to-mouth and a fraction $1 - \lambda$ is Ricardian. The point of this extension is not to reproduce the full richness of heterogeneous-agent macroeconomics. The point is to break the representative-agent logic in the minimal way needed to show that not all households respond to monetary or fiscal policy through the same channel.

In RANK, the household side is summarized by one Euler equation, so aggregate demand moves mainly through intertemporal substitution. In TANK, by contrast, only part of the household sector smooths consumption intertemporally. The remaining households consume their current disposable income period by period. This simple split is enough to make transfers, labor income, and the distribution of current income matter for aggregate demand.

10.1 Economic structure of TANK

The TANK model divides households into two groups.

- (i) **Ricardian households.** These households have access to the asset market, choose consumption intertemporally, and satisfy an Euler equation.
- (ii) **Hand-to-mouth households.** These households do not use the asset market to smooth consumption. Their consumption is pinned down by current disposable income.

The parameter λ denotes the population share of hand-to-mouth households. The remaining share $1 - \lambda$ consists of Ricardian households.

This split should be interpreted as a reduced-form way of capturing limited asset-market participation or severe liquidity constraints. It is deliberately simple. The model does not yet generate an endogenous wealth distribution or a rich cross-section of marginal propensities to consume. But it does allow one to distinguish households that react strongly to current income from households that react directly to the real interest rate.

10.2 Hand-to-mouth households

Hand-to-mouth households consume their current disposable income:

$$C_t^H = w_t N_t^H + T_t^H. \quad (111)$$

They do not save or borrow, so their budget constraint binds period by period.

Equation (111) is best understood as a behavioral restriction rather than as the solution to a standard intertemporal optimization problem. The household is assumed not to smooth consumption

through the asset market. As a result, any change in current labor income or transfers feeds directly into current consumption.

This is the central source of non-Ricardian behavior in TANK. If current wages fall, or transfers are reduced, hand-to-mouth consumption falls immediately. If transfers rise, hand-to-mouth consumption rises one for one.

In many textbook TANK models, hand-to-mouth households also supply labor. One can either assume that they choose labor optimally, as in Galí-type TANK setups, or keep labor income reduced-form. For transparency, it is useful to allow labor supply and consumption behavior to differ conceptually: the key non-Ricardian feature is not labor supply, but the absence of intertemporal asset smoothing.

10.3 Ricardian households

Ricardian households solve a standard intertemporal optimization problem. They choose sequences $\{C_t^R, B_{t+1}^R, N_t^R\}_{t \geq 0}$ to maximize

$$\max_{\{C_t^R, B_{t+1}^R, N_t^R\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_t^R)^{1-\sigma} - 1}{1-\sigma} - \chi \frac{(N_t^R)^{1+\phi_n}}{1+\phi_n} \right] \quad (112)$$

subject to

$$C_t^R + B_{t+1}^R = R_t B_t^R + w_t N_t^R + T_t^R. \quad (113)$$

Here T_t^R denotes the net transfers received by Ricardian households. Depending on the fiscal closure of the model, this object may include lump-sum transfers, taxes, and profit income.

The associated first-order conditions are the usual ones:

$$(C_t^R)^{-\sigma} = \lambda_t^R, \quad (114)$$

$$\chi (N_t^R)^{\phi_n} = \lambda_t^R w_t, \quad (115)$$

$$\lambda_t^R = \beta \mathbb{E}_t [R_{t+1} \lambda_{t+1}^R]. \quad (116)$$

Combining them yields the Ricardian Euler equation

$$(C_t^R)^{-\sigma} = \beta \mathbb{E}_t [R_{t+1} (C_{t+1}^R)^{-\sigma}], \quad (117)$$

and the intratemporal labor-supply condition

$$w_t = \chi (N_t^R)^{\phi_n} (C_t^R)^{\sigma}. \quad (118)$$

Thus Ricardian households behave exactly like the representative household in RANK. The difference is that they now make up only part of the population.

10.4 Aggregation

Aggregate consumption is the population-weighted average of the two groups:

$$C_t = \lambda C_t^H + (1 - \lambda) C_t^R. \quad (119)$$

Similarly, aggregate labor is

$$N_t = \lambda N_t^H + (1 - \lambda) N_t^R. \quad (120)$$

If only Ricardian households participate in asset markets, then aggregate bond holdings satisfy

$$(1 - \lambda) B_t^R = B_t^g, \quad (121)$$

where B_t^g denotes the supply of government debt or, more generally, the aggregate safe asset held by the private sector.

This bond-market-clearing condition is important. It implies that Ricardians absorb the entire stock of assets. Hand-to-mouth households have zero net asset positions by assumption. As a result, movements in interest income accrue only to the Ricardian group.

10.5 A simple resource view

If the baseline economy has no capital and no government purchases, aggregate output must equal aggregate consumption:

$$Y_t = C_t. \quad (122)$$

In that case, combining (119) with goods-market clearing gives

$$Y_t = \lambda C_t^H + (1 - \lambda) C_t^R. \quad (123)$$

This expression already highlights the key conceptual difference from RANK. Aggregate demand is no longer tied to a single Euler equation. Instead, it is the weighted sum of one Euler-driven consumption component and one current-income-driven consumption component.

10.6 Interpretation

TANK makes visible two distinct channels of monetary transmission.

1. Intertemporal substitution. Ricardian households respond directly to the real interest rate. When monetary policy raises the real rate, they postpone consumption and save more, just as in RANK.

2. Current-income effects. Hand-to-mouth households react mainly to current disposable income. Their consumption depends on wages, employment, taxes, and transfers in the current period. They do not smooth these shocks through the asset market.

This distinction matters immediately for policy analysis. In RANK, a contractionary monetary shock reduces demand mainly because the representative household chooses to postpone consumption. In TANK, that mechanism still operates for Ricardians, but hand-to-mouth households also react through movements in labor income and transfers. Aggregate demand therefore depends not only on the real rate, but also on the distribution of current income across household types.

10.7 How TANK modifies the New Keynesian backbone

The nominal side of the economy remains the same as in RANK. Firms still face monopolistic competition and Calvo price rigidity. Monetary policy still follows a Taylor rule. The difference lies in the household block and therefore in the transmission of aggregate demand.

In RANK, the dynamic IS relation comes directly from the Euler equation of the representative household. In TANK, there is no longer a single household Euler equation that describes aggregate consumption. Instead, aggregate demand combines:

- (i) one forward-looking component, coming from Ricardian consumption;
- (ii) one current-income component, coming from hand-to-mouth consumption.

This is the key conceptual contribution of TANK. The model preserves the familiar New Keynesian nominal backbone, but weakens the idea that all demand adjustment is governed by one intertemporal condition.

10.8 A useful reduced-form representation

Although the exact aggregate demand equation depends on the full closure of the model, the qualitative logic is easy to summarize. A contractionary monetary shock affects the two groups differently.

For Ricardians, a higher real interest rate lowers current consumption through intertemporal substitution:

$$\downarrow C_t^R \quad \text{when} \quad \uparrow (i_t - \mathbb{E}_t \pi_{t+1}).$$

For hand-to-mouth households, the effect works mainly through current labor income and transfers:

$$\downarrow C_t^H \quad \text{when} \quad \downarrow w_t N_t^H \text{ or } \downarrow T_t^H.$$

Aggregate consumption therefore moves according to

$$\Delta C_t = \lambda \Delta C_t^H + (1 - \lambda) \Delta C_t^R,$$

so the overall demand response depends both on the size of the hand-to-mouth sector and on the incidence of the shock across groups.

10.9 Why TANK matters

TANK is often the first model in which students see clearly that heterogeneity can matter even when it is introduced in a very stylized way.

It matters for at least three reasons.

- (i) It breaks the tight link between aggregate consumption and a single Euler equation.
- (ii) It makes fiscal incidence relevant: who receives transfers or pays taxes now matters for aggregate demand.

- (iii) It creates a primitive but economically meaningful notion of inequality dynamics, because the two groups' consumptions can move differently over the cycle.

For these reasons, TANK is much more than “RANK plus one extra household.” It is the first model in the sequence in which aggregate transmission depends explicitly on the composition of the household sector.

10.10 What TANK still misses

At the same time, TANK remains highly stylized. It does not generate an endogenous wealth distribution, does not model precautionary saving, and does not allow households to move through a rich state space. The distinction between Ricardian and hand-to-mouth households is imposed exogenously rather than arising from incomplete markets and individual histories.

These limitations are not defects of the model's purpose. They are exactly what make TANK useful pedagogically. It is the simplest environment in which one can introduce heterogeneous consumption responses without yet leaving the world of analytically tractable New Keynesian models.

10.11 Figures for TANK

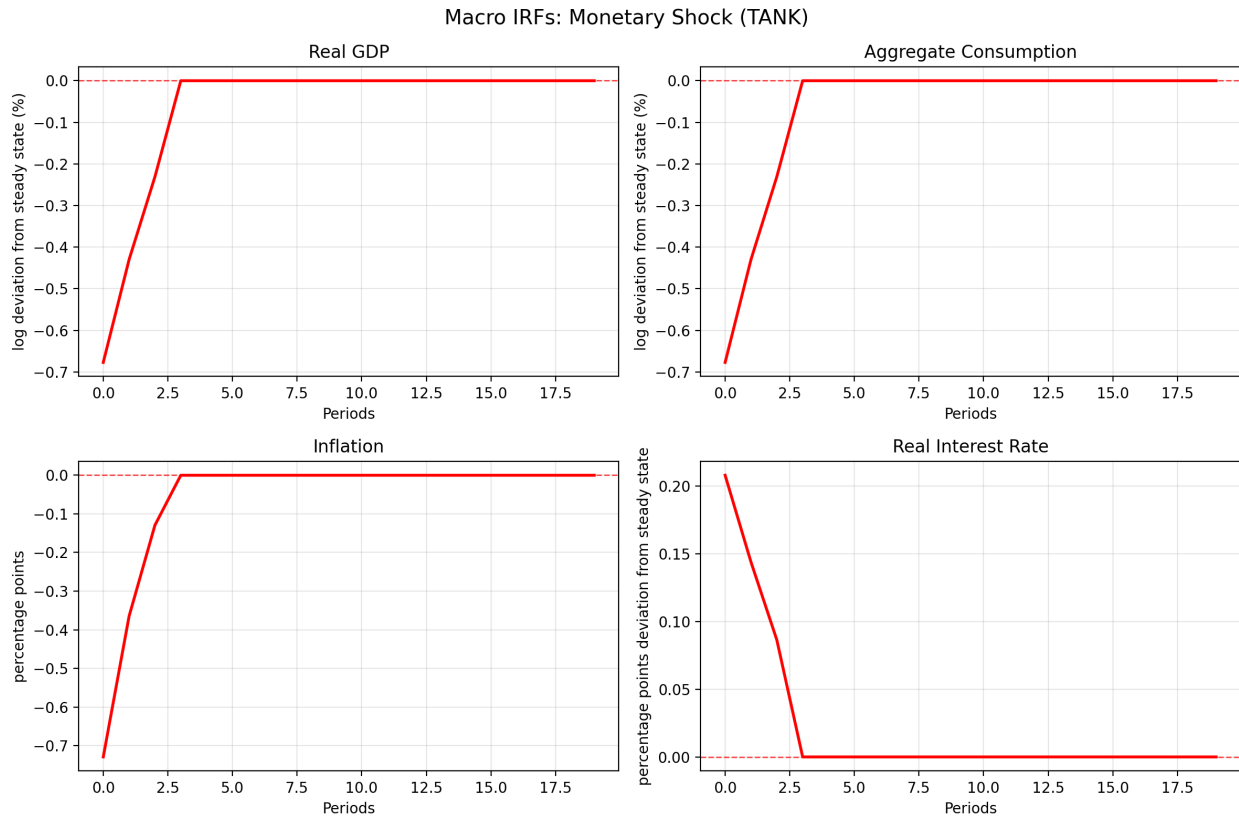


Figure 3: TANK macro impulse responses to a contractionary monetary shock.

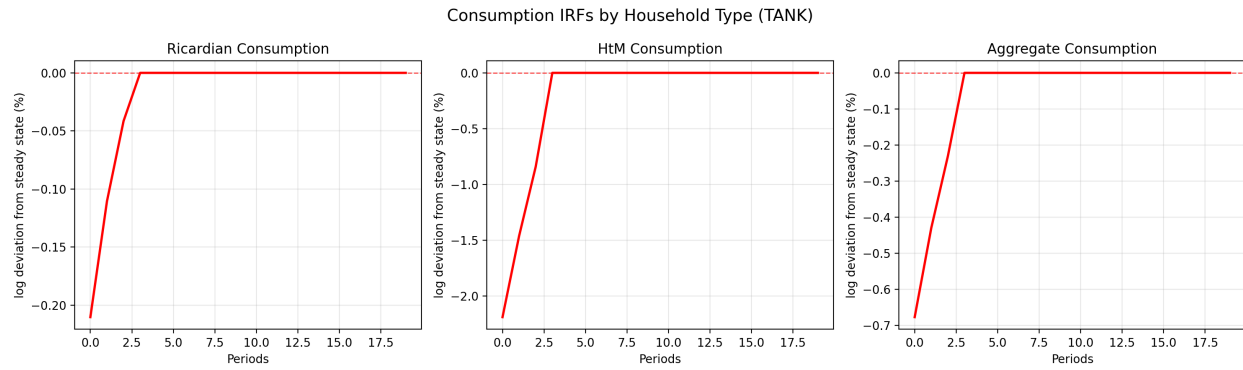


Figure 4: TANK consumption responses by household type.

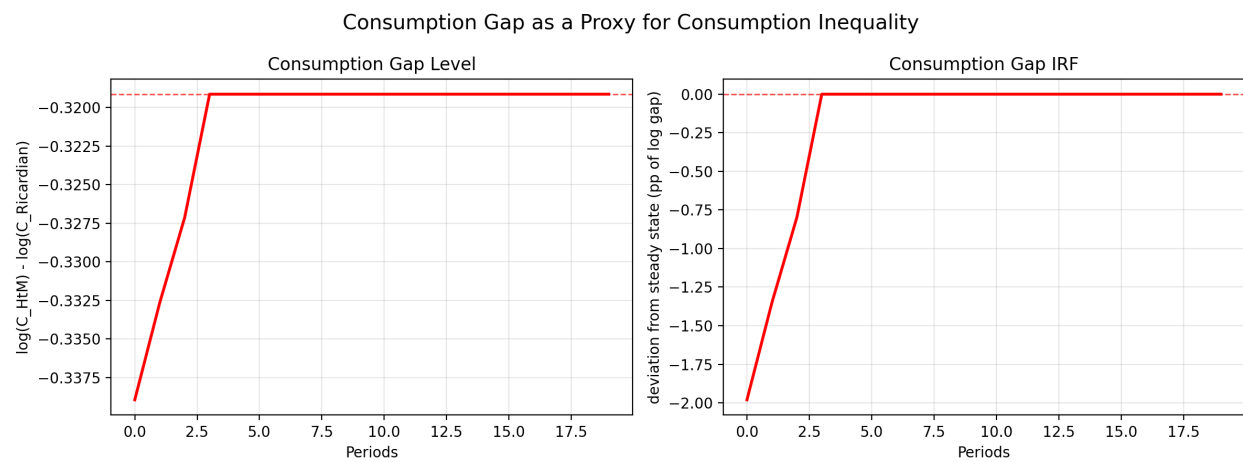


Figure 5: TANK consumption gap as a proxy for consumption inequality.

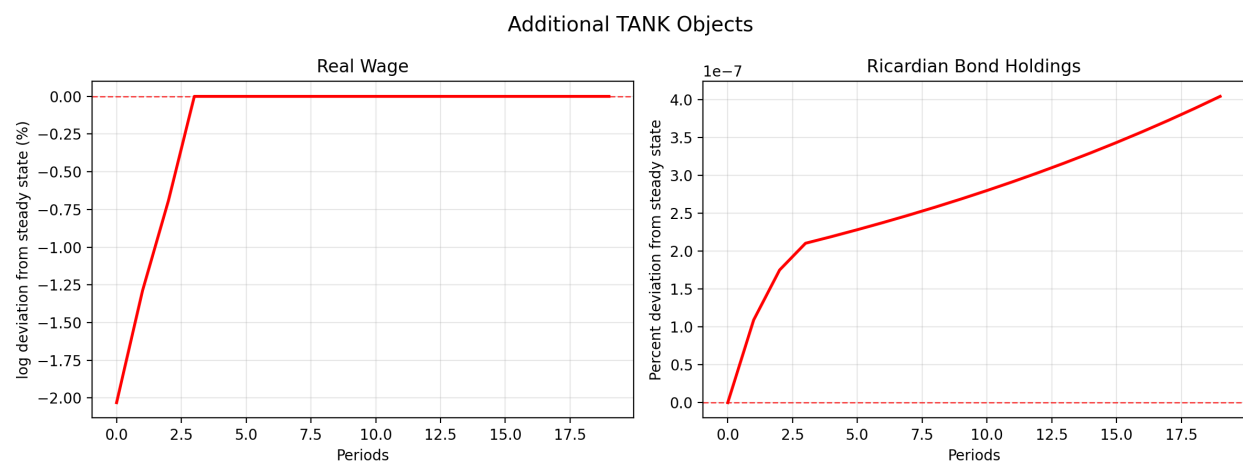


Figure 6: Additional TANK transition objects.

10.12 Reading the figures

The aggregate panel shows that the New Keynesian backbone remains recognizable. Output, inflation, and the interest rate still move in a way that resembles the representative-agent benchmark because the firm block and policy block are unchanged.

The household-type panel reveals the main new mechanism. Consumption no longer moves through a single margin. Ricardian households adjust consumption through the real interest rate, while hand-to-mouth households move mainly with current disposable income. This difference is the essential reason why TANK can generate larger and more realistic roles for transfers, taxes, and labor-income incidence.

The consumption-gap figure makes visible a rudimentary form of inequality dynamics. Even in a model with only two household groups, the cycle redistributes consumption responses across households. This is one of the key conceptual bridges from RANK to richer heterogeneous-agent environments.

The additional transition objects help the reader inspect the moving parts behind the headline impulse responses. They are especially useful for checking whether the relative movements of wages, labor income, transfers, and group-level consumption match the logic of the model.

11 THANK: tractable heterogeneous-agent New Keynesian model

This section introduces THANK as an intermediate step between TANK and HANK. The goal is not simply to add another household type. The goal is to introduce genuinely dynamic heterogeneity — risk, state transitions, and self-insurance — while keeping the model analytically transparent enough that the reader can still see the main mechanisms without immediately solving a full high-dimensional heterogeneous-agent problem.

In this sense, THANK should be read as a *tractable heterogeneous-agent New Keynesian model*. It is richer than TANK because households can move across states over time and therefore care about future constraints in a forward-looking way. But it remains much simpler than HANK because the household sector is summarized by a small number of states rather than by a full continuous distribution of assets and idiosyncratic income.

11.1 What THANK adds beyond TANK

The main point of THANK is not merely to have two household types. TANK already does that. What THANK adds is a dynamic state structure that allows the model to capture several mechanisms that are central in quantitative HANK.

- (i) **State transitions.** Households can move between an unconstrained state and a constrained state.
- (ii) **Precautionary saving.** Unconstrained households save partly because they may become constrained in the future.
- (iii) **Cyclical inequality.** The relative disposable incomes of constrained and unconstrained households can vary over the business cycle.

- (iv) **Cyclical risk.** The probability of falling into the constrained state may itself depend on aggregate conditions.
- (v) **Richer MPC logic.** Aggregate demand depends not only on intertemporal substitution, but also on the distribution of households across states and on the changing value of liquidity.

This is the key conceptual move beyond TANK. In TANK, type heterogeneity is static and exogenously fixed. In THANK, heterogeneity becomes dynamic: households care not only about who they are today, but also about the probability of becoming constrained tomorrow.

11.2 States and timing

Consider a population of households that can be in one of two states in each period:

- (i) state S , the saver or unconstrained state;
- (ii) state H , the constrained or hand-to-mouth state.

A household in state S has access to liquid bonds and can carry liquidity into the next period. A household in state H is constrained and consumes its current disposable income. Let λ_t denote the measure of households in state H , so that $1 - \lambda_t$ is the measure in state S .

State transitions are governed by a time-varying Markov matrix

$$\Pi_t = \begin{pmatrix} p_{SS,t} & p_{SH,t} \\ p_{HS,t} & p_{HH,t} \end{pmatrix}, \quad (124)$$

where each row sums to one. For example, $p_{SH,t}$ is the probability that a household currently in the saver state enters the constrained state next period.

The law of motion for the constrained share is therefore

$$\lambda_{t+1} = (1 - \lambda_t)p_{SH,t} + \lambda_t p_{HH,t}. \quad (125)$$

This equation is important because it makes the cross-sectional composition of the household sector endogenous over time. Even with only two states, the model already contains a distributional state variable: the fraction of households currently constrained.

11.3 Constrained-state behavior

A household in state H consumes current disposable income:

$$C_t^H = Y_t^{disp,H}. \quad (126)$$

Let disposable income in the constrained state be

$$Y_t^{disp,H} = w_t N_t^H + T_t^H - \tau_t^H + \omega_t^H, \quad (127)$$

where $w_t N_t^H$ is labor income, T_t^H denotes transfers, τ_t^H taxes, and ω_t^H any additional income component relevant for the constrained household, such as transfers tied to policy or state-contingent support.

Equation (126) should again be interpreted as a reduced-form liquidity constraint. The constrained household does not solve a standard intertemporal saving problem in the current period. Its consumption is pinned down by current disposable resources. This implies a high marginal propensity to consume out of current income.

The constrained state therefore plays a similar role to hand-to-mouth status in TANK, but with one crucial difference: here households can move into and out of that state. The constrained state is not a fixed type. It is a realizable condition.

11.4 Saver-state behavior

A saver chooses next period's liquid asset position in order to smooth consumption and self-insure against future state transitions. Let b denote beginning-of-period liquid asset holdings for a household currently in state S . The value function of a saver is

$$V_t^S(b) = \max_{b' \geq \underline{b}} \left\{ u(C_t^S) + \beta \left[p_{SS,t} \mathbb{E}_t V_{t+1}^S(b') + p_{SH,t} \mathbb{E}_t V_{t+1}^H(b') \right] \right\}, \quad (128)$$

subject to

$$C_t^S + b' = R_t b + Y_t^{disp,S}, \quad (129)$$

where

$$Y_t^{disp,S} = w_t N_t^S + \Pi_t^S / P_t + T_t^S - \tau_t^S. \quad (130)$$

The borrowing limit is $\underline{b}$. The key new term relative to TANK is the continuation value:

$$p_{SS,t} \mathbb{E}_t V_{t+1}^S(b') + p_{SH,t} \mathbb{E}_t V_{t+1}^H(b').$$

Saving today is valuable not only because it supports future consumption if the household remains unconstrained, but also because it provides liquidity if the household falls into the constrained state tomorrow.

This is the source of precautionary saving in THANK.

11.5 The saver Euler condition

Proposition 11.1. *The saver-state Euler condition is*

$$u_c(C_t^S) = \beta \left[p_{SS,t} \mathbb{E}_t \left[R_{t+1} u_c(C_{t+1}^S) \right] + p_{SH,t} \mathbb{E}_t V_{b,t+1}^H(b') \right]. \quad (131)$$

Proof. Differentiate the Bellman equation (128) with respect to b' . The marginal cost of choosing a larger b' is that current consumption falls one for one, generating marginal utility loss $u_c(C_t^S)$. The marginal benefit is the discounted expected continuation value of extra liquidity tomorrow. If the household remains in state S , that extra unit yields the usual marginal value of wealth in the unconstrained state. If instead the household transitions to state H , the same extra unit is valued according to the derivative of the constrained-state value function with respect to available liquid resources. Weighting these continuation values by the corresponding transition probabilities yields (131). $\square$

To make the logic clearer, one may write the Euler condition in a more compact form:

$$u_c(C_t^S) = \beta \mathbb{E}_t \left[\mathcal{M}_{t+1}^{THANK} \right], \quad (132)$$

where the effective continuation term $\mathcal{M}_{t+1}^{THANK}$ depends both on tomorrow's real return and on tomorrow's state. Unlike in RANK or even TANK, the relevant continuation value is not summarized by one common marginal utility term alone. It depends on the possibility of entering a constrained state in which liquidity is especially valuable.

11.6 Interpretation: precautionary saving and the value of liquidity

The saver Euler condition should be interpreted carefully. In RANK, saving is valuable because it transfers resources across time. In THANK, saving is valuable for two reasons.

First, it still performs the usual intertemporal smoothing role. Second, it also provides self-insurance against the possibility of becoming constrained in the future. This second motive is the essence of precautionary saving.

An extra unit of liquid wealth tomorrow is particularly valuable in bad states. If the household falls into the constrained state, that unit may relax the effective tightness of its liquidity position or raise the resources available while constrained. Therefore, the marginal value of wealth is state-dependent. That state dependence is what makes THANK a tractable bridge toward HANK.

11.7 Cyclical inequality

One simple way to summarize distributional movements in the model is through the relative disposable incomes of saver and constrained households. Define the income-inequality measure

$$\mathcal{I}_t = \log \left(\frac{Y_t^{disp,S}}{Y_t^{disp,H}} \right). \quad (133)$$

If $\mathcal{I}_t$ rises in recessions, inequality is countercyclical: constrained households become relatively poorer precisely when aggregate conditions deteriorate. This matters for aggregate demand because constrained households tend to have higher MPCs. A redistribution of disposable income away from them therefore has stronger contractionary effects than an equal decline in average income spread uniformly across households.

THANK is therefore able to capture a basic version of cyclical inequality dynamics without requiring a full wealth distribution.

11.8 Cyclical risk

A second key mechanism is cyclical transition risk. Let the probability of entering the constrained state depend on aggregate conditions:

$$p_{SH,t} = \bar{p}_{SH} + \psi_y \hat{y}_t + \varepsilon_t^p. \quad (134)$$

If $\psi_y < 0$, then weak aggregate activity raises the probability of falling into the constrained state. In that case recessions are periods not only of low current income, but also of heightened future risk.

This matters because policy affects more than current demand. It also affects households' perception of future risk. A contractionary shock can reduce output today and, by worsening transition risk, induce additional precautionary behavior among saver households. This creates an amplification channel that is absent in TANK.

11.9 Aggregation

Aggregate consumption is

$$C_t = \lambda_t C_t^H + (1 - \lambda_t) C_t^S. \quad (135)$$

Because the state shares evolve endogenously over time, aggregate consumption depends not only on each state's consumption level but also on the distribution of households across states.

If liquid public debt is positive, asset-market clearing is

$$(1 - \lambda_t) b_t^S + \lambda_t b_t^H = B_t^g. \quad (136)$$

In the simplest formulation one may set

$$b_t^H = 0,$$

so that only unconstrained households hold liquid assets:

$$(1 - \lambda_t) b_t^S = B_t^g. \quad (137)$$

This closure makes clear that the constrained share of the population affects not only aggregate consumption but also the distribution of asset holdings.

11.10 THANK and the aggregate Euler logic

A useful reduced-form summary of the THANK demand block is

$$\hat{c}_t = \mu_0 \hat{y}_t + \mu_1 \mathbb{E}_t[\hat{c}_{t+1}] - \mu_2 \hat{r}_t + \mu_3 \hat{\mathcal{I}}_t + \mu_4 \widehat{p_{SH,t}}. \quad (138)$$

This expression should not be read as a primitive structural equation, but as a pedagogical summary of the aggregate logic. It highlights that aggregate demand depends not only on the real interest rate and expected future demand, but also on cyclical inequality and cyclical risk.

Each term has an interpretation.

- (i) $\mu_1 \mathbb{E}_t[\hat{c}_{t+1}]$ captures the forward-looking component inherited from saver households.
- (ii) $-\mu_2 \hat{r}_t$ captures the intertemporal-substitution channel.
- (iii) $\mu_3 \hat{\mathcal{I}}_t$ captures distributional shifts in disposable income across high- and low-MPC states.
- (iv) $\mu_4 \widehat{p_{SH,t}}$ captures changing transition risk and the associated precautionary motive.

Compared with RANK and TANK, the aggregate Euler logic is therefore richer. The economy no longer behaves as if aggregate consumption were chosen by a single household, nor as if the only heterogeneity were a fixed split between asset holders and non-asset holders.

11.11 THANK and the New Keynesian cross

The nominal side of the model remains the standard New Keynesian one. Firms still generate a Phillips curve through Calvo pricing, and monetary policy still operates through a Taylor rule. What changes is the household block and therefore the determination of aggregate demand.

The reduced-form system can be summarized schematically as follows:

$$\hat{c}_t = \mu_0 \hat{y}_t + \mu_1 \mathbb{E}_t[\hat{c}_{t+1}] - \mu_2 \hat{r}_t + \mu_3 \hat{\mathcal{I}}_t + \mu_4 \widehat{p_{SH,t}}, \quad (139)$$

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa \hat{m}c_t, \quad (140)$$

$$\hat{i}_t = \phi_\pi \pi_t + \phi_y \hat{y}_t + \nu_t. \quad (141)$$

This system makes the point clearly. Inflation and monetary policy still obey the familiar New Keynesian backbone. But aggregate demand now depends on a richer set of household-side state variables than in either RANK or TANK.

11.12 THANK versus TANK

THANK differs from TANK in three essential ways.

- (i) **Type is no longer static.** In TANK, a household is Ricardian or hand-to-mouth by fixed label. In THANK, state is dynamic and households can move between states over time.
- (ii) **Saving reflects risk.** In TANK, the forward-looking group saves only for standard intertemporal-smoothing reasons. In THANK, saver behavior also reflects precautionary motives.
- (iii) **Macro transmission depends on risk and composition.** In TANK, aggregate demand depends on the weighted sum of two fixed groups. In THANK, it also depends on the endogenous evolution of state shares and on the cyclicalities of transition risk.

This is why THANK should not be interpreted as simply “a more elaborate TANK.” It is the first model in the sequence in which households care explicitly about future constraint risk.

11.13 THANK versus HANK

THANK remains much simpler than HANK. It does not track a continuous wealth distribution, and it does not require a full global solution for policy functions over a rich asset-income state space. But it captures several HANK-like channels in a compact form.

- (i) Risk and self-insurance.
- (ii) Heterogeneous MPCs and iMPCs.
- (iii) Cyclical inequality.
- (iv) Fiscal incidence.
- (v) A richer aggregate Euler logic than TANK.

The point is not that THANK replaces HANK. It does not. The point is that THANK allows the reader to understand, in a controlled analytical setting, why the richer household-side mechanisms emphasized in HANK matter for New Keynesian transmission.

11.14 Why THANK is pedagogically valuable

THANK is especially useful in a manual like this one because it provides a genuine bridge. TANK shows that heterogeneous disposable-income responses matter. HANK shows that incomplete markets, precautionary saving, and the wealth distribution matter. THANK sits in between by preserving a small-state structure while introducing exactly the channels that TANK is missing.

For students, this is often the first model in which the phrase *tractable HANK* becomes meaningful. One can see how risk, state transitions, and liquidity valuation enter the aggregate demand block without yet confronting the full machinery of a continuous-distribution heterogeneous-agent model.

12 HANK: heterogeneous-agent New Keynesian model

We now move to the full baseline HANK environment. This is the point at which household heterogeneity is no longer represented by a small number of fixed groups or a small number of states alone. Instead, the household sector is described by policy functions defined over an asset-income state space and by the full cross-sectional distribution of households across those states.

This is the defining conceptual step from THANK to HANK. In RANK, the household sector collapses to one Euler equation. In TANK, it becomes a weighted average of two groups. In THANK, it becomes a small-state system with transition risk. In HANK, by contrast, aggregate dynamics depend on the joint evolution of individual decisions and the distribution of households. The model therefore combines New Keynesian nominal rigidities with incomplete markets, idiosyncratic risk, borrowing constraints, and endogenous wealth heterogeneity.

12.1 Why HANK is qualitatively different

HANK is not simply THANK with more states. The key difference is structural.

In THANK, one can still summarize the household sector with a small number of states and a relatively compact set of transition equations. In HANK, household heterogeneity is represented through a continuous or finely discretized distribution over asset holdings and idiosyncratic income states. Each household solves a dynamic savings problem under incomplete markets, and aggregate outcomes depend on the interaction between:

- (i) individual policy functions for consumption and savings;
- (ii) the law of motion of the cross-sectional distribution;
- (iii) market-clearing conditions that feed back into prices and returns.

This means that the object replacing the representative-agent Euler equation is no longer a single aggregate condition. Instead, one must keep track of how many households are constrained, how many are wealthy, how many are exposed to labor-income risk, how those households respond to

shocks, and how the resulting distributional shifts feed back into aggregate demand and asset-market clearing.

That is why HANK is a genuinely different environment rather than just a richer calibration of earlier models.

12.2 Economic environment

The baseline HANK household faces three sources of complexity absent from RANK.

First, labor income is idiosyncratic: households differ in their current productivity or earnings state. Second, markets are incomplete: households cannot insure perfectly against that idiosyncratic risk. Third, borrowing is limited: households face a lower bound on asset positions. These three ingredients jointly generate precautionary saving, a nondegenerate wealth distribution, and large variation in marginal propensities to consume.

The nominal side of the model remains New Keynesian. Firms are monopolistically competitive, prices are sticky, and monetary policy follows an interest-rate rule. The household side, however, is now rich enough that monetary policy affects the economy not only through intertemporal substitution, but also through income incidence, liquidity, self-insurance, and redistribution across balance sheets.

12.3 States and budget constraint

Each household enters period t with individual state (a_t, z_t) , where $a_t \in \mathcal{A}$ denotes current asset holdings and $z_t \in \mathcal{Z}$ denotes an idiosyncratic labor-income or productivity state. The process for z_t is exogenous and typically modeled as a finite-state Markov chain:

$$z_{t+1} \sim \Pi(z_t, \cdot), \quad (142)$$

or equivalently,

$$\Pr(z_{t+1} = z' \mid z_t = z) = \Pi(z, z'). \quad (143)$$

The household budget constraint is

$$c_t + a_{t+1} = R_t a_t + w_t z_t + T_t - \tau_t, \quad (144)$$

subject to

$$a_{t+1} \geq \underline{a}. \quad (145)$$

Here R_t is the gross real return on liquid assets carried into period t , $w_t z_t$ is idiosyncratic labor income, and $T_t - \tau_t$ denotes net transfers. The lower bound $\underline{a}$ captures the limited ability of households to borrow. If $\underline{a} = 0$, households cannot borrow at all. If $\underline{a} < 0$, they may borrow up to a finite limit.

This borrowing constraint is central. It implies that low-asset households cannot fully smooth shocks through the asset market. As a result, those households typically exhibit high marginal propensities to consume out of current income.

12.4 Bellman problem

Given aggregate prices and policies, the household solves a standard incomplete-markets dynamic programming problem. The Bellman equation is

$$V_t(a, z) = \max_{a' \geq \underline{a}} \left\{ \frac{c^{1-\sigma} - 1}{1 - \sigma} + \beta \sum_{z' \in \mathcal{Z}} \Pi(z, z') V_{t+1}(a', z') \right\}, \quad (146)$$

with

$$c = R_t a + w_t z + T_t - \tau_t - a'. \quad (147)$$

This problem has the standard dynamic-programming interpretation. The household chooses next period's asset position a' to trade off current consumption against future self-insurance. Saving more today lowers current consumption, but provides resources tomorrow in states where labor income may be low. Saving less today raises current consumption, but leaves the household more exposed to future income risk and to the borrowing constraint.

The presence of both idiosyncratic risk and borrowing limits is what generates precautionary behavior and a nontrivial wealth distribution.

12.5 Euler equation and borrowing constraint

For households that are interior, meaning the borrowing constraint does not bind, the Euler equation is

$$c_t(a, z)^{-\sigma} = \beta R_{t+1} \sum_{z' \in \mathcal{Z}} \Pi(z, z') c_{t+1}(a', z')^{-\sigma}. \quad (148)$$

This has the same formal structure as a standard Euler equation, but its interpretation is different from RANK. The household's marginal utility tomorrow depends on the full continuation problem under idiosyncratic risk. Since households differ in both assets and income states, there is no single aggregate marginal utility summarizing the entire economy.

If the borrowing constraint binds, then the Euler equation is replaced by the Euler inequality:

$$c_t(a, z)^{-\sigma} \geq \beta R_{t+1} \sum_{z' \in \mathcal{Z}} \Pi(z, z') c_{t+1}(a', z')^{-\sigma}, \quad a_{t+1} = \underline{a}. \quad (149)$$

This inequality is important economically. Constrained households would like to borrow more against future income, but cannot. As a result, their current consumption is pinned down much more tightly by current resources. These are precisely the households that tend to display high MPCs and that make monetary and fiscal policy more powerful through distributional channels.

12.6 Policy functions

The solution to the household problem is summarized by policy functions

$$a' = g_t(a, z), \quad c = c_t(a, z). \quad (150)$$

These functions tell us what each household does as a function of its current state. A wealthy household with high assets and good income prospects may choose to save substantially. A poor

household near the borrowing limit may consume nearly all of its current income. A household in a bad income state may decumulate assets to smooth consumption.

The policy functions are therefore highly state-dependent. This is one of the defining differences from RANK and TANK. In those models, the household block can be summarized by a small number of aggregate equations. In HANK, the household block is summarized instead by a pair of functions defined over the individual state space.

12.7 The cross-sectional distribution

Let $\mu_t(a, z)$ denote the cross-sectional distribution of households over assets and idiosyncratic states at date t . This distribution is the second central object of the HANK model, alongside the policy functions.

Given the policy rule $g_t(a, z)$ and the transition matrix for idiosyncratic income, the distribution evolves forward according to a law of motion of the form

$$\mu_{t+1} = \Gamma_t(\mu_t, g_t), \quad (151)$$

where Γ_t is the operator that pushes the distribution forward through households' saving decisions and idiosyncratic state transitions.

This law of motion is conceptually simple but economically profound. It says that aggregate dynamics depend not only on what an individual household would do in a given state, but also on how many households are currently in each state. The same policy rule can generate very different aggregate outcomes depending on the distribution to which it is applied.

12.8 Aggregation

Aggregate consumption is obtained by integrating individual consumption decisions over the distribution:

$$C_t = \sum_{z \in \mathcal{Z}} \int_{\mathcal{A}} c_t(a, z) \mu_t(a, z) da. \quad (152)$$

Aggregate asset demand is

$$A_t = \sum_{z \in \mathcal{Z}} \int_{\mathcal{A}} g_t(a, z) \mu_t(a, z) da. \quad (153)$$

If the only net asset in the economy is government debt, asset-market clearing requires

$$A_t = B_t^g. \quad (154)$$

These aggregation equations are conceptually straightforward but central to HANK. The aggregate economy is no longer built from a representative household. It is built by integrating heterogeneous decisions over a nondegenerate distribution.

12.9 Why the distribution matters

In HANK, the real interest rate must clear the asset market given the saving desires of the entire distribution. This is the defining difference relative to all previous models.

In RANK, there is one household, so there is no nontrivial distribution. In TANK, there are two groups, so the household side is summarized by a few group-level equations. In THANK, one tracks a small number of states and state shares. In HANK, by contrast, the distribution itself becomes a state variable for the aggregate economy.

This matters because two economies with the same aggregate asset holdings can react very differently to the same policy shock if their distributions differ. An economy in which many households are close to the borrowing constraint will generally display stronger consumption responses to income shocks than an economy in which wealth is concentrated among unconstrained households with low MPCs.

Thus, in HANK, the distribution is not merely descriptive. It is causal.

12.10 Stationary equilibrium and steady-state distribution

In steady state, aggregate prices are constant and the distribution is stationary. A stationary distribution $\bar{\mu}(a, z)$ satisfies the household policy rules and reproduces itself under the induced law of motion:

$$\bar{\mu} = \Gamma(\bar{\mu}, \bar{g}). \quad (155)$$

A steady-state HANK equilibrium therefore consists not only of constant aggregate quantities and prices, but also of a fixed cross-sectional wealth-income distribution consistent with household optimization and market clearing. This is another fundamental difference from RANK. In HANK, computing the steady state already requires solving a nontrivial fixed-point problem involving both household behavior and the induced distribution.

12.11 Transmission channels

HANK decomposes monetary transmission into several channels.

- (i) **Intertemporal substitution.** Higher real interest rates induce some households to postpone consumption.
- (ii) **Labor-income effects.** Monetary policy affects employment, wages, and therefore current income across the distribution.
- (iii) **Profit and transfer incidence.** Changes in profits, taxes, and transfers affect households differently depending on ownership and policy design.
- (iv) **Precautionary-saving responses.** Changes in risk and in the value of self-insurance alter desired savings.
- (v) **Redistribution across heterogeneous balance sheets.** Households differ in net asset positions, liquidity, and exposure to nominal and real returns, so policy redistributes resources across balance sheets.

The great contribution of the HANK literature is to show that monetary transmission is not exhausted by the representative-agent real-rate channel. Aggregate demand moves because policy changes the whole distribution of income and resources, and because households at different points in the distribution respond very differently.

12.12 A useful comparison with earlier models

It is worth summarizing the logic of the sequence.

In RANK, monetary policy works mainly through one Euler equation. In TANK, it works through one Euler equation plus one current-income-driven consumption block. In THANK, it works through state transitions, precautionary motives, and the evolution of state shares. In HANK, it works through the interaction of policy functions, borrowing constraints, idiosyncratic risk, and the entire distribution of households.

This is why HANK is the natural end point of the progression developed in this manual. It contains the representative-agent intertemporal channel, the TANK-style income-incidence channel, the THANK-style risk and self-insurance channel, and an additional distributional layer that cannot be compressed into a small number of states without losing important economic structure.

12.13 Why HANK is computationally harder

The richness of HANK comes at a computational cost. Solving the model requires more than deriving a few first-order conditions.

One must:

- (i) solve the household dynamic-programming problem;
- (ii) compute policy functions over the state space;
- (iii) find the stationary distribution implied by those policy rules;
- (iv) aggregate over that distribution;
- (v) and, for dynamics, track how the distribution evolves after shocks.

This is the main reason why HANK is harder than all previous models in the sequence. The gain is realism and richer transmission. The cost is that the household sector can no longer be summarized by a compact analytical block alone.

12.14 Figures for HANK

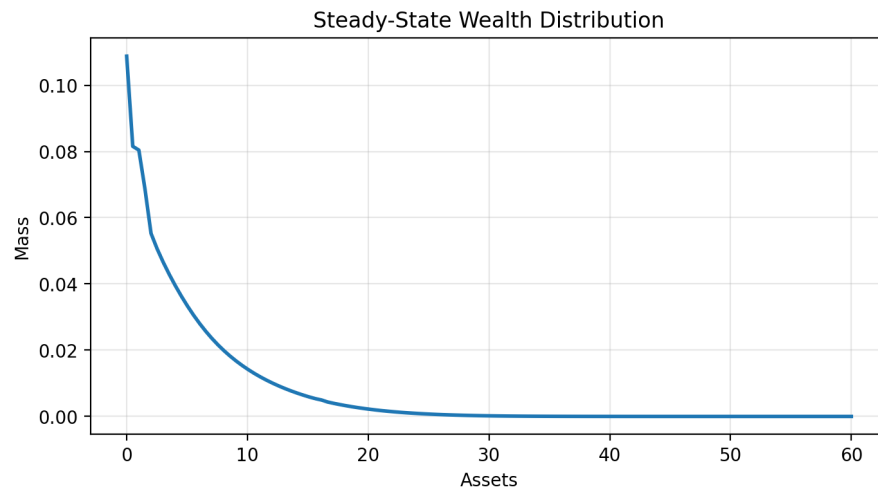


Figure 7: HANK steady-state wealth distribution.

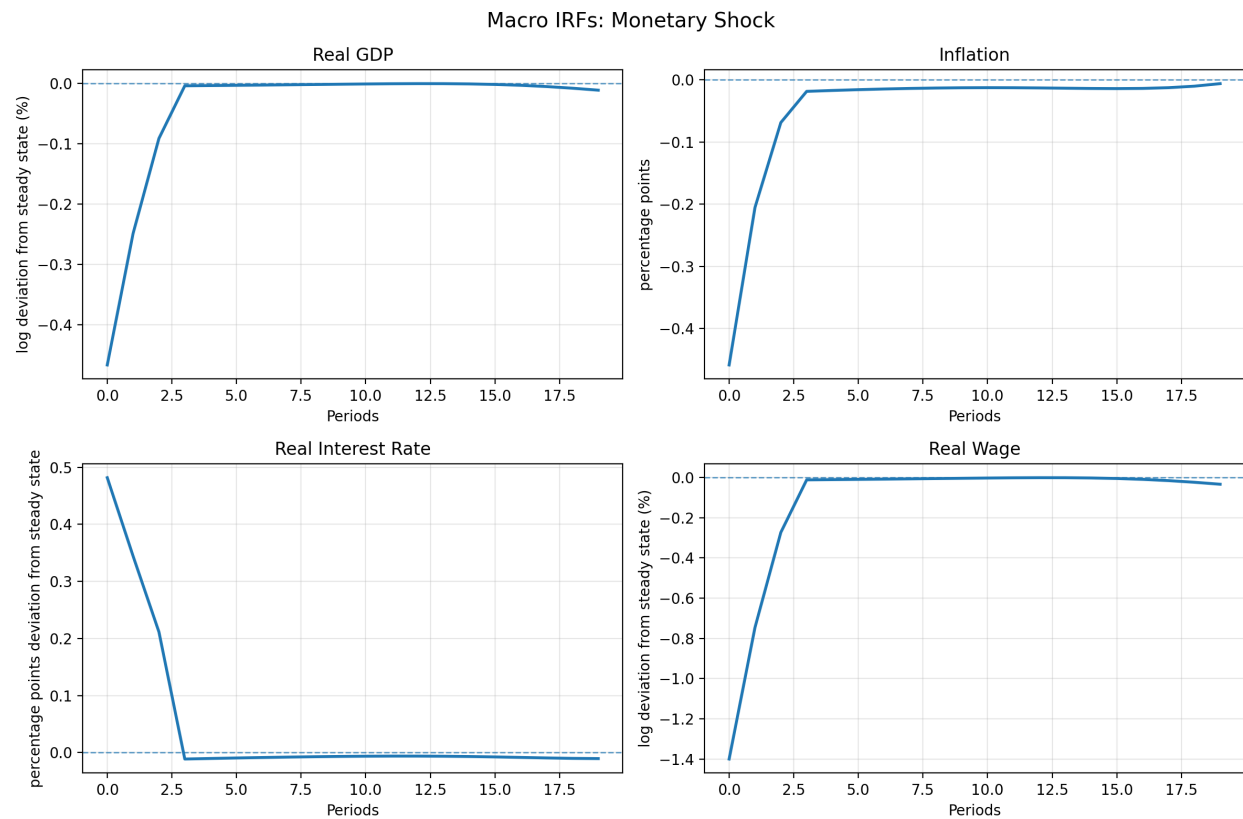


Figure 8: HANK macro impulse responses to a contractionary monetary shock.

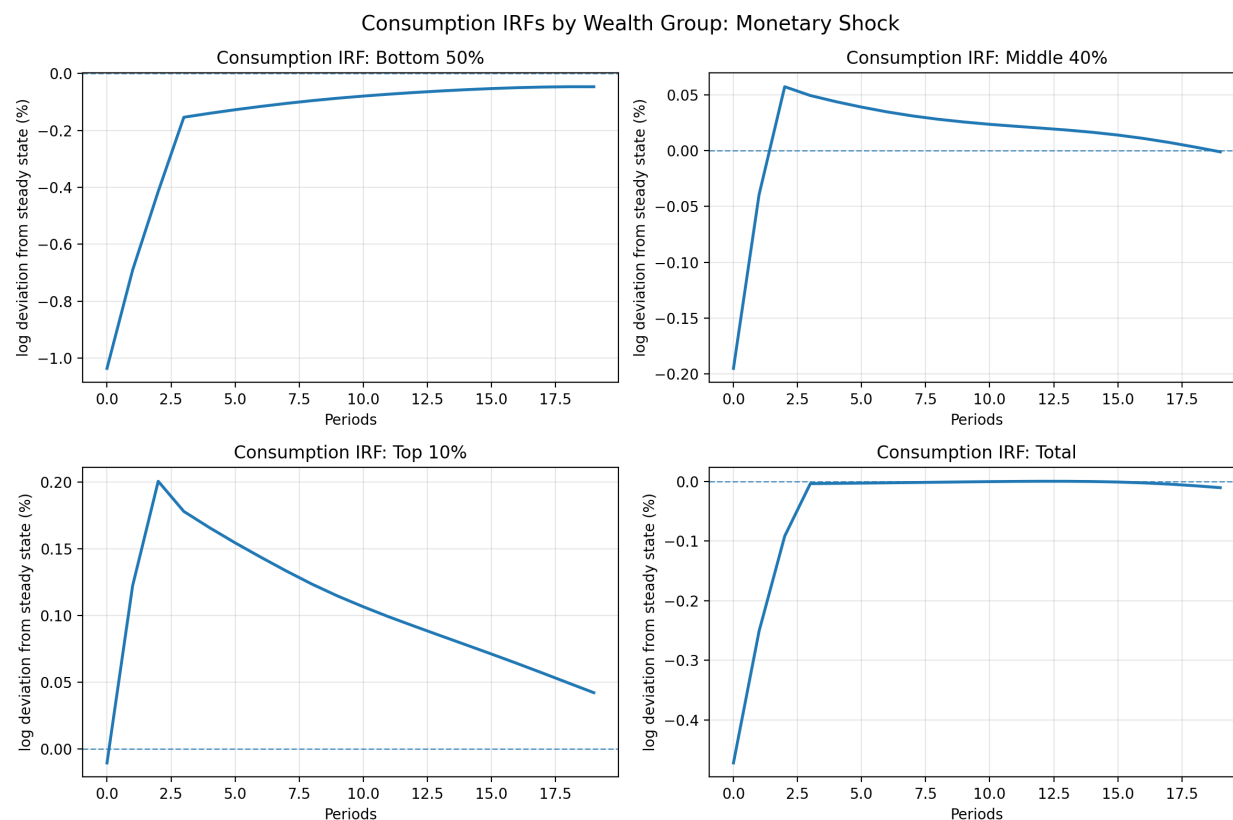


Figure 9: HANK consumption responses by wealth group.

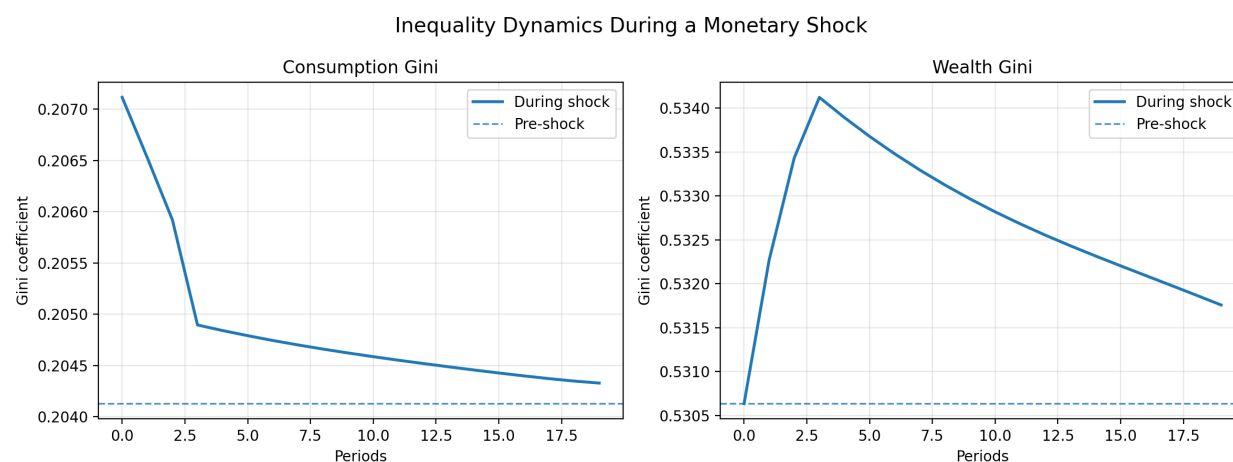


Figure 10: HANK consumption and wealth Gini coefficients during the monetary shock.

12.15 Reading the figures

The wealth-distribution figure shows the central state object absent from RANK, TANK, and THANK. In HANK, one cannot understand aggregate transmission without understanding where households sit in the wealth distribution.

The macro impulse responses may still display the familiar New Keynesian sign pattern: a

contractionary monetary shock raises the real interest rate, lowers output, reduces labor income, and pushes inflation down. But the underlying mechanism is richer than in earlier models because the aggregate response is the sum of very different household-level responses.

The group-specific consumption figure is especially informative. It shows that households in different parts of the wealth distribution do not react equally. Low-wealth or constrained households typically move consumption more strongly because current income and liquidity matter more for them. Wealthier households may respond more through intertemporal substitution and balance-sheet revaluation.

The Gini paths emphasize that monetary policy is not distribution-neutral. Even when aggregate variables move in familiar ways, the shock can change both consumption inequality and wealth inequality along the transition. This is one of the central reasons HANK matters for modern monetary macroeconomics.

13 Wrapping up: Comparing RANK, TANK, THANK, and HANK

Across all four models, the New Keynesian nominal backbone is the same. What changes is the internal logic of the household side.

13.1 Comparison by mechanism

- (i) In RANK, everything runs through one Euler equation and one labor-supply condition.
- (ii) In TANK, part of demand is Euler-equation driven and part is current-income driven.
- (iii) In THANK, demand is shaped by current income, transitions across states, self-insurance, precautionary saving, cyclical inequality, and risk.
- (iv) In HANK, demand is shaped by the whole joint distribution of assets and earnings.

13.2 Comparison by tractability

- (i) RANK is fully tractable.
- (ii) TANK is nearly as tractable.
- (iii) THANK is tractable in the sense that key HANK channels can often be represented analytically or with a small state space.
- (iv) HANK is computationally rich but analytically much less transparent.

Table 2: Practical comparison

Model	Main strength	Main weakness	Best use case
RANK	Clean aggregate logic	No distributional structure	Introductory NK analysis and benchmark policy intuition
TANK	Minimal heterogeneity with low cost	No precautionary saving or transitions	Teaching heterogeneous consumption responses and fiscal incidence
THANK	Captures risk and self-insurance in a tractable way	No full wealth distribution	Bridge from TANK to HANK; cyclical inequality, iMPCs, and amplification
HANK	Rich distributional and aggregate transmission	High computational burden	Research questions about inequality, balance sheets, policy incidence, and micro-consistent dynamics

14 Wrapping up: From equations to computation

The transition from theory to code is where many students first get lost. The key is to treat each model as a sequence of objects to be computed.

14.1 RANK

- (i) Choose parameters.
- (ii) Solve for the deterministic steady state.
- (iii) Linearize or solve the nonlinear transition path.
- (iv) Feed in shocks.
- (v) Check Euler and market-clearing residuals.

14.2 TANK

- (i) Choose the hand-to-mouth share.
- (ii) Solve the Ricardian block.
- (iii) Recover hand-to-mouth consumption from current income.
- (iv) Aggregate.
- (v) Check goods and asset markets.

14.3 THANK

- (i) Choose transition probabilities between states.
- (ii) Solve the saver block taking transition risk into account.
- (iii) Recover constrained-state consumption.
- (iv) Compute aggregate consumption using state shares.
- (v) If liquidity is positive, solve for bond demand and market clearing.
- (vi) Track how inequality and risk vary with aggregate conditions.

14.4 HANK

- (i) Guess aggregate prices.
- (ii) Solve the household dynamic program.
- (iii) Compute policy functions.
- (iv) Compute the invariant distribution.
- (v) Aggregate.
- (vi) Update prices until markets clear.

14.5 Sequence-space methods

Modern HANK work often uses sequence-space Jacobians. The basic idea is to compute how aggregates respond marginally to sequences of shocks around the steady state and then solve a linear system over time. This is one of the major computational breakthroughs in contemporary heterogeneous-agent macroeconomics [3].

15 Extensions: richer fiscal policy and labor-market structure

The baseline models developed in this manual deliberately keep the fiscal and labor-market blocks as simple as possible. This is a pedagogical choice. A simple fiscal block helps isolate the household-side mechanisms that distinguish RANK, TANK, THANK, and HANK. A simple labor block helps keep the pricing side transparent and prevents the model from becoming overloaded too early.

But this simplification has limits. Many substantive applications in modern macroeconomics require a richer treatment of fiscal policy, labor-market institutions, and wage setting. Once household heterogeneity matters, the composition of taxes, transfers, labor income, and debt ownership becomes central to aggregate transmission. Similarly, once labor income is an important transmission margin, a richer labor-market structure — especially one with wage rigidity — becomes hard to avoid.

This section therefore sketches two natural extensions of the baseline framework: a more structured fiscal block and a richer labor market with differentiated labor, union wage setting, and sticky wages.

15.1 A richer fiscal block

Let the government choose public spending G_t , taxes τ_t , transfers $\mathcal{T}_t$, and one-period nominal debt $B_{g,t+1}$ subject to the nominal budget constraint

$$P_t G_t + \mathcal{T}_t + (1 + i_{t-1})B_{g,t} = \tau_t + B_{g,t+1}. \quad (156)$$

This equation states that government uses — purchases of goods, transfers, and repayment of previously issued debt with nominal interest — must be financed by current taxes and newly issued debt.

Dividing through by the price level gives the real budget constraint

$$G_t + \frac{\mathcal{T}_t}{P_t} + R_{t-1} \frac{B_{g,t}}{P_t} = \frac{\tau_t}{P_t} + \frac{B_{g,t+1}}{P_t}, \quad (157)$$

where

$$R_{t-1} = \frac{1 + i_{t-1}}{1 + \pi_t}$$

is the ex post gross real return on nominal government debt carried from $t - 1$ to t .

Equivalently, one may write

$$G_t + \frac{\mathcal{T}_t}{P_t} + R_{t-1} \frac{B_{g,t}}{P_t} = \frac{\tau_t}{P_t} + \frac{B_{g,t+1}}{P_t}. \quad (158)$$

The aggregate resource constraint becomes

$$Y_t = C_t + G_t, \quad (159)$$

abstracting from capital accumulation, investment adjustment costs, and other real uses of output. Thus public spending absorbs part of total output and changes the mapping from aggregate production to private consumption.

A simple debt-stabilizing fiscal rule can be written as

$$\frac{\tau_t}{P_t} = \bar{\tau} + \phi_b \left(\frac{B_{g,t}}{P_{t-1}} - \bar{b} \right) + \phi_y (Y_t - \bar{Y}), \quad (160)$$

where $\bar{\tau}$ is steady-state real tax revenue, $\bar{b}$ is the target debt level in real terms, and $\phi_b > 0$ measures the responsiveness of taxes to deviations of debt from target.

This rule is only one possible closure. One could instead allow debt stabilization to occur through spending, through transfers, through state-contingent taxes, or through a combination of all three. In heterogeneous-agent settings, this choice matters enormously because the macroeconomic effects of fiscal adjustment depend on who pays and who receives.

15.2 Why richer fiscal structure matters

In representative-agent models, fiscal composition is often less central than the aggregate path of deficits and taxes. In heterogeneous-agent models, however, fiscal composition becomes a first-order issue.

The reason is simple: households differ in MPCs, balance sheets, labor-income exposure, and tax incidence. A transfer directed toward constrained households has a different aggregate effect from the same transfer directed toward wealthy savers. A tax levied on labor income has different distributional consequences from a tax levied on asset income. Public debt matters not only as an intertemporal government financing device, but also as a private-sector asset held unequally across households.

A richer fiscal block therefore matters for at least four reasons.

- (i) **Transfer incidence.** Transfers directed toward high-MPC households generate stronger demand effects.
- (ii) **Tax incidence.** Different tax instruments alter disposable income differently across the distribution.
- (iii) **Debt ownership.** Interest payments on government debt accrue to whoever holds that debt, which makes debt service distributionally relevant.
- (iv) **Interaction with monetary policy.** Inflation and interest-rate movements change the real value of nominal debt and therefore interact with fiscal redistribution.

These issues are present even in TANK, become richer in THANK, and become central in HANK.

15.3 Differentiated labor and labor aggregation

The baseline models above treat labor in a reduced-form way. A richer structure assumes that labor services are differentiated across workers or unions and then aggregated into a composite labor input by a competitive labor packer.

Let labor services be aggregated according to the CES index

$$N_t = \left(\int_0^1 N_t(j)^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dj \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}, \quad (161)$$

where $\varepsilon_w > 1$ is the elasticity of substitution across labor varieties.

Given the aggregate labor requirement N_t , the labor packer minimizes total wage payments

$$\min_{\{N_t(j)\}_{j \in [0,1]}} \int_0^1 W_t(j) N_t(j) dj \quad (162)$$

subject to (161). Exactly as in the final-good aggregation problem on the firm side, cost minimization yields labor demand for variety j :

$$N_t(j) = \left(\frac{W_t(j)}{W_t} \right)^{-\varepsilon_w} N_t, \quad (163)$$

where the aggregate wage index is

$$W_t = \left(\int_0^1 W_t(j)^{1-\varepsilon_w} dj \right)^{\frac{1}{1-\varepsilon_w}}. \quad (164)$$

This block mirrors the goods side exactly. Differentiated labor implies that each worker type or union faces a downward-sloping demand curve for its own labor variety. That, in turn, creates wage-setting power and allows wage rigidity to matter in a parallel way to price rigidity.

15.4 Union wage setting

A standard interpretation is that labor variety j is supplied by a union or household type that sets its nominal wage subject to labor demand (163). The union chooses its wage taking into account both the utility consequences for the supplying households and the demand consequences of charging a higher relative wage.

Under flexible wages, the optimal wage can be expressed as a markup over the marginal rate of substitution between consumption and labor. Under sticky wages, by contrast, only a subset of unions can reset wages each period, exactly as in Calvo pricing on the goods side.

If wages are Calvo-sticky, only a fraction $1 - \theta_w$ of wage setters can reset wages each period. Let π_t^w denote wage inflation. Log-linearization then yields a wage Phillips curve of the form

$$\pi_t^w = \beta \mathbb{E}_t[\pi_{t+1}^w] - \kappa_w \hat{\mu}_{w,t}, \quad (165)$$

where $\hat{\mu}_{w,t}$ is the wage markup gap, that is, the deviation of the actual wage markup from its efficient or flexible-wage benchmark.

The exact expression for κ_w depends on the Calvo parameter θ_w , the elasticity of substitution across labor varieties, and the structure of household labor supply. But the economic meaning is already clear: wages adjust sluggishly, and the incentive to reset them depends on how far current wages are from the wage path consistent with optimal labor-market conditions.

15.5 Why richer labor structure matters

Sticky wages matter for at least four reasons.

- (i) **They add another nominal rigidity.** Inflation dynamics are no longer governed solely by price stickiness.
- (ii) **They alter marginal cost dynamics.** Since wages are a key component of firms' costs, sluggish wage adjustment changes the behavior of real marginal cost and therefore price inflation.
- (iii) **They change labor-income incidence.** When wages adjust slowly, monetary and fiscal shocks affect labor income differently across households and over time.
- (iv) **They enrich transmission in heterogeneous-agent models.** In HANK-type settings, labor income is often one of the main channels through which policy reaches high-MPC households, so wage-setting institutions become especially important.

In short, once one cares about distributional transmission, the labor market cannot be treated as a mere side block. Wage rigidity, union structure, and labor-income heterogeneity become integral parts of the macroeconomic mechanism.

15.6 How these extensions connect to the earlier models

These richer fiscal and labor blocks do not replace the logic of RANK, TANK, THANK, or HANK. They enrich it.

In RANK, they add realism and broaden the set of policy questions that can be asked. In TANK, they sharpen the role of fiscal incidence and labor-income heterogeneity. In THANK, they interact with transition risk, self-insurance, and cyclical inequality. In HANK, they become central because the effects of policy depend on who receives transfers, who pays taxes, who owns debt, and whose labor income is most affected by wage dynamics.

These extensions therefore belong naturally at the end of the manual. They show how the baseline sequence can be opened toward the frontier of research without abandoning the logic developed earlier.

16 Conclusion

This manual has presented a route through New Keynesian models with progressively richer household heterogeneity. The purpose of the progression from RANK to TANK to THANK to HANK is not to claim that one model is always better than another. The purpose is to understand what each model adds, what each model suppresses, and when each model is the appropriate tool for the question at hand.

Each model in the sequence has its own pedagogical and analytical role.

- (i) **RANK** is the benchmark for clean aggregate logic. It isolates the standard New Keynesian interaction between the Euler equation, the Phillips curve, and monetary policy.
- (ii) **TANK** is the benchmark for minimal heterogeneity with high pedagogical payoff. It shows immediately that disposable-income incidence and MPC heterogeneity matter for aggregate demand.
- (iii) **THANK** is the benchmark for introducing self-insurance, precautionary saving, cyclical inequality, transition risk, and richer iMPCs without yet moving to the full distributional machinery of HANK.
- (iv) **HANK** is the benchmark for genuine distributional macroeconomics under incomplete markets, where aggregate outcomes depend on policy functions, borrowing constraints, and the entire cross-sectional distribution.

The sequence matters because each step changes not only the realism of the model, but also the economic interpretation of policy transmission. In RANK, monetary policy works mainly through one intertemporal substitution margin. In TANK, it also works through current-income incidence. In THANK, it works through transition risk and self-insurance. In HANK, it works through the full interaction of balance sheets, heterogeneous MPCs, incomplete markets, and the endogenous distribution of households.

For students, the most important lesson is not merely that richer models produce different impulse responses. The deeper lesson is that richer models explain *why* those responses occur, *which*

households drive them, and *through which margins* policy is transmitted across the income and wealth distribution.

That is the ultimate rationale for the progression developed in this manual. One does not move from RANK to HANK because complexity is fashionable. One moves because different questions require different household structures, and because modern macroeconomics increasingly needs models capable of connecting aggregate stabilization to distributional incidence.

The best way to read these models, therefore, is not as competing tribes. It is to read them as a nested family. Each model preserves part of the architecture of the previous one while relaxing a simplification that has become too restrictive for the question under study. Seen in that way, the route from RANK to TANK to THANK to HANK is not a sequence of disconnected frameworks. It is a coherent map of modern New Keynesian macroeconomics.

A Notation guide

Symbol	Meaning
C_t	Aggregate consumption
C_t^R	Ricardian consumption in TANK
C_t^H	Hand-to-mouth consumption in TANK or constrained-state consumption in THANK notation
C_t^S	Saver-state consumption in THANK
Y_t	Final output
G_t	Government spending
N_t	Aggregate labor input
$P_t(i)$	Price of variety i
P_t	Aggregate price index
π_t	Price inflation
π_t^w	Wage inflation
i_t	Nominal interest rate
R_t	Gross ex-post real interest rate
r_t	Net real interest rate
w_t	Real wage
$W_t(j)$	Nominal wage of labor type j
mc_t	Real marginal cost
ε_p	Elasticity of substitution across goods
ε_w	Elasticity of substitution across labor varieties
θ_p	Calvo price stickiness parameter
θ_w	Calvo wage stickiness parameter
ν_t	Monetary policy shock
B_g	Government supply of safe assets in simple baseline environments
$B_{g,t+1}$	Endogenous public debt in richer fiscal block
τ_t	Taxes
$\mathcal{T}_t$	Aggregate transfers

T_t	Net transfers
a_t	Individual asset holdings in HANK
b_t	Liquid asset holdings in THANK notation
z_t	Idiosyncratic productivity state in HANK
Π	Markov transition matrix for idiosyncratic states
λ	Share of hand-to-mouth households in TANK
λ_t	Share of constrained households in THANK
σ	Coefficient of relative risk aversion / inverse intertemporal elasticity
ϕ_n	Inverse Frisch elasticity parameter
ϕ_π	Monetary policy response to inflation
ϕ_y	Monetary policy response to activity
ϕ_b	Fiscal response to debt
MPC	Marginal propensity to consume
iMPC	Intertemporal marginal propensity to consume

B Timing conventions

Households enter period t with beginning-of-period assets B_t in RANK and TANK, with liquid assets b_t in THANK, or with assets a_t in HANK. They receive labor income, transfers, profits, or taxes during the period. They then choose the next-period asset position.

Inflation between t and $t + 1$ is denoted by π_{t+1} in the exact gross return formula

$$R_t = \frac{1 + i_t}{1 + \pi_{t+1}}. \quad (166)$$

C A derivation of the reduced-form labor-supply relation

Suppose the household has period utility

$$u(C_t, N_t) = \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \chi \frac{N_t^{1+\phi_n}}{1+\phi_n}. \quad (167)$$

The intratemporal condition is

$$u_c(C_t, N_t)w_t = -u_n(C_t, N_t). \quad (168)$$

Since

$$u_c(C_t, N_t) = C_t^{-\sigma}, \quad -u_n(C_t, N_t) = \chi N_t^{\phi_n}, \quad (169)$$

we obtain

$$C_t^{-\sigma} w_t = \chi N_t^{\phi_n}, \quad (170)$$

or

$$w_t = \chi N_t^{\phi_n} C_t^{\sigma}. \quad (171)$$

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